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Study of Decoherence Effects on Polarization Entanglement Distri- bution through Optical Fibers

TESI DI LAUREA MAGISTRALE IN
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Abstract

As quantum technologies progress toward distributed architectures, optical quantum links are becoming increasingly important for both interconnecting remote quantum processing nodes and enabling secure quantum communication. Polarization-encoded photonic qubits are particularly suited to fiber-based transmission, since polarization can be prepared, manipulated, and measured using well-established optical components.

Among the applications considered in this thesis, particular attention is devoted to entanglement-based quantum key distribution, with the BBM92 protocol providing the main operational reference. In this context, the preservation of high-quality entanglement is directly related to the suitability of the distributed state for secure quantum communication. The *Quantum Bit Error Rate* (QBER) provides a direct protocol-level criterion for assessing whether the measured correlations are compatible with secure key extraction, while the *Secure Key Rate* (SKR) represents the corresponding operational throughput once experimental parameters and security quality are taken into account.

This thesis develops a progressive and unified framework for studying the propagation of polarization-entangled photon pairs through optical fibers, connecting physical descriptions commonly treated separately at different levels of approximation. The main objective is to relate fiber and source parameters to the quality of the distributed entanglement and to the operational requirements of BBM92-based communication. To this purpose, the framework connects the physical parameters of the optical link with state-quality metrics and with protocol-relevant quantities derived from the polarization correlations, particularly the QBER. This provides a practical procedure for identifying operating regions of the physical parameter space in which the predicted polarization correlations satisfy an adopted BBM92 QBER criterion, without assuming a universal concurrence threshold.

The modeling hierarchy starts from the residual polarization evolution after partial compensation. A fixed relative phase first distinguishes coherent changes in measured correlations from genuine entanglement degradation. The model is then extended to unresolved phase fluctuations through the coherence parameter μ , and to independent depolariz-

ing effects in the two fiber arms through depolarizing parameters p_A and p_B . Finally, these effective descriptions are connected to polarization-mode dispersion (PMD), where finite-bandwidth photons undergo frequency-dependent birefringent transformations and polarization mixedness emerges after tracing over unresolved spectral degrees of freedom.

The framework is implemented in a modular MATLAB simulation environment that integrates quantum-state preparation, fiber-channel modeling, statistical averaging, spectral tracing, and polarization-correlation analysis. Analytical and numerical results are used to study how source properties, fiber parameters, PMD, and compensation conditions affect the surviving entanglement.

The theoretical and numerical analysis is complemented by a preliminary laboratory study of a simplified two-arm, BBM92-oriented polarization platform. The experimental activity is designed to characterize, under controlled conditions, the polarization transformations and degradation mechanisms introduced by the optical link and by selected fiber components. Although the present setup does not directly implement a complete entanglement-based QKD protocol, it provides a first experimental step toward assessing the practical requirements of fiber-based entanglement distribution and toward establishing the connection between fiber-induced polarization degradation and, in a future complete BBM92 implementation, experimentally measurable quantities such as the QBER and the Secure Key Rate.

Keywords: Polarization Entanglement, Optical Fibers, BBM92, Quantum Key Distribution, Secure Key Rate, Quantum Channels, Decoherence, Depolarization.

Abstract in lingua italiana

DA INSERIRE QUANDO L'ABSTRACT È DEFINITIVO

Parole chiave: Entanglement di Polarizzazione, Fibre Ottiche, BBM92, Canali Quantistici, Decoerenza, Depolarizzazione.

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Introduction

As quantum computing technologies continue to advance and quantum communication links become increasingly relevant for interconnecting remote quantum systems, the reliable transmission of quantum states between distant nodes is becoming an increasingly important requirement in information technology. Optical fibers represent a natural transmission medium for many such applications because of their compatibility with existing telecommunications infrastructure and their suitability for long-distance photonic links [1, 2].

Among the different photonic degrees of freedom, polarization provides a particularly convenient encoding for quantum information. Polarization-entangled photon pairs can be generated, manipulated, and measured using established optical components, making them attractive resources for entanglement distribution and entanglement-based quantum communication protocols [3]. Their transmission through real optical fibers, however, is affected by birefringence, polarization rotations, phase accumulation, temporal effects, and environmental fluctuations [4]. While deterministic local transformations can remain coherent and preserve entanglement, unresolved fluctuations or coupling to additional degrees of freedom can produce effective decoherence and degrade the polarization correlations accessible at the receiver [5, 6].

The preservation of polarization entanglement is particularly relevant to entanglement-based quantum key distribution. In the BBM92 protocol, an entangled photon pair is distributed between two distant parties, Alice and Bob, who independently choose between alternative polarization measurement bases to establish a secure key [3]. Fiber propagation and environmental perturbations can degrade the distributed correlations, increasing the quantum bit error rate (QBER) and ultimately limiting the suitability of the link for secure quantum communication.

BBM92 therefore provides the main operational reference for this thesis, whose objective is to investigate how source properties and fiber propagation conditions affect the quality of distributed polarization entanglement and its suitability for quantum communication. The reference configuration consists of an entangled-photon source connected to two spa-

tially separated fiber arms, A and B , with one photon propagating through each arm and polarization used as the qubit degree of freedom. The work progressively connects the physical mechanisms affecting this configuration with their mathematical modeling, numerical simulation, and operational consequences.

Chapter 1 establishes the physical and mathematical framework required for this analysis. It introduces the description of polarization qubits and bipartite states, the density-operator formalism, reduced states, local measurements, and polarization correlations. The chapter also presents the quantities used throughout the thesis to characterize state quality, coherence, entanglement. Particular attention is devoted to concurrence as the principal entanglement measure adopted in the numerical analysis. A brief description of the BBM92 protocol and of its main quantities of interest is also provided, establishing the connection between the quantum-state description and the operational perspective used in the subsequent chapters.

Chapter 2 describes how optical fiber propagation modifies the polarization state through a progressive hierarchy of models with increasing physical detail. A central distinction is made between coherent polarization evolution, which can modify measured correlations without necessarily reducing purity or entanglement, and effective decoherence arising from unresolved fluctuations or coupling to unobserved degrees of freedom. Reduced models are first used to isolate coherent phase evolution, statistical dephasing, and local depolarizing effects in the two propagation arms, providing analytically interpretable descriptions of the different degradation mechanisms. The analysis is then extended to a frequency-dependent treatment of polarization-mode dispersion (PMD) [7–9], connecting fiber birefringence and the spectral properties of the photon pair to the effective degradation of polarization entanglement. The resulting hierarchy therefore provides a continuous path from simplified effective channels to a physical description of frequency-dependent fiber propagation.

Chapter 3 describes the implementation of the developed models in a modular MATLAB simulation framework. It investigates through numerical experiments how source properties, fiber parameters, PMD, and compensation conditions affect the distributed entanglement. The analysis aims to identify regions of parameter space in which the distributed state satisfies selected state-quality and BBM92-oriented operational criteria, assess the validity of reduced models against the physical frequency-dependent description, and relate the resulting entanglement degradation to protocol-relevant quantities such as the QBER.

Chapter 4 complements the theoretical and numerical work with a preliminary laboratory

study of polarization-entanglement distribution. A simplified two-arm platform based on entangled-photon propagation, polarization control, single-photon detection, and coincidence measurements is considered. The objective of this activity is to provide a first laboratory approach to the measurement conditions and practical requirements relevant to BBM92-oriented quantum communication over optical fibers. At the present stage, the experimental activity is therefore intended as an exploration of the conditions of propagation in the fiber links rather than as a complete BBM92 implementation.

Finally, Chapter ?? summarizes the main theoretical, numerical, and experimental results and discusses the possible extensions of the developed framework. Particular attention is given to the connection between physical fiber-channel modeling, the numerical identification of entanglement-quality regions, and their operational interpretation in the context of entanglement-based quantum communication.

1 | Mathematical and Physical Framework

This chapter introduces the physical representation and mathematical tools used throughout the thesis to describe polarization-entangled photon pairs and their evolution through optical-fiber channels. The treatment is formulated within the density-operator framework of quantum information theory, which provides a common description for pure states, statistical mixtures, local measurements, and noisy quantum evolution [10].

The chapter first defines the bipartite polarization system and the reference Bell state. The density-operator and reduced-state formalisms are then introduced, followed by the description of two-photon polarization measurements and correlations. The main quantities used to characterize the propagated quantum state are subsequently defined, with particular attention to purity, fidelity, concurrence, tangle, and entanglement of formation.

Finally, the chapter introduces the BBM92 entanglement-based quantum key distribution protocol as the main operational reference of this work. Its measurement scheme and principal quantities of interest, including the quantum bit error rate and secret-key conditions, are presented in order to establish the connection between the quality of the distributed entangled state and its suitability for secure quantum communication. This provides the operational basis for interpreting the fiber-propagation models and numerical results developed in the following chapters.

1.1. Bipartite Polarization System

The physical system considered in this work consists of a pair of polarization-entangled photons propagating along two spatially separated optical paths, denoted by A and B . In the experimental configuration considered later in this thesis, the photon pairs are produced through high-fidelity spontaneous parametric down-conversion (SPDC).

The quantum information carried by each photon is encoded in its polarization degree of freedom. Horizontal and vertical polarization are identified with the computational basis

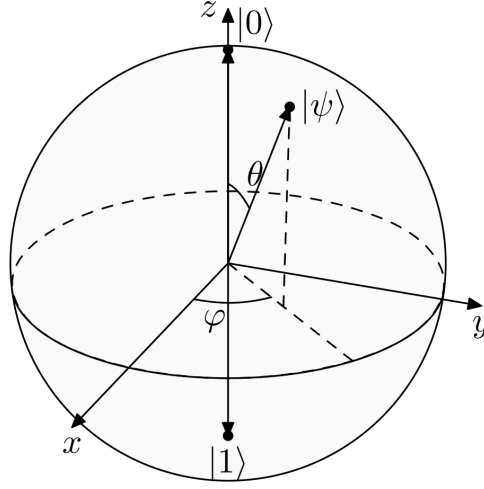


Figure 1.1: Bloch-sphere representation of a single polarization qubit. The horizontal and vertical polarization states correspond to $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$.

according to

$$|H\rangle \equiv |0\rangle, \quad |V\rangle \equiv |1\rangle. \quad (1.1)$$

Each photon therefore defines a two-dimensional Hilbert space,

$$\mathcal{H}_A \simeq \mathbb{C}^2, \quad \mathcal{H}_B \simeq \mathbb{C}^2. \quad (1.2)$$

The Bloch sphere provides a geometric representation of the state of a single qubit, as illustrated in Fig. 1.1. Pure states are represented by points on the surface of the sphere, while the polar angle and azimuthal angle determine the relative amplitudes and phase of the two basis-state components. Coherent superpositions occupy the remaining points on the sphere, with equal-weight superpositions lying on the equator.

The joint system is described in the tensor-product Hilbert space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \simeq \mathbb{C}^2 \otimes \mathbb{C}^2, \quad (1.3)$$

which has dimension four.

The ordered computational basis adopted throughout the thesis is

$$\mathcal{B}_{AB}^{01} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}, \quad (1.4)$$

where the first entry refers to subsystem A and the second to subsystem B . In polarization notation, the same basis is

$$\mathcal{B}_{AB}^{HV} = \{|HH\rangle, |HV\rangle, |VH\rangle, |VV\rangle\}. \quad (1.5)$$

Consequently, pure states of the complete system are represented by four-component complex vectors and operators acting on both photons by 4×4 matrices. This basis convention is used consistently in the analytical derivations and in the MATLAB implementation.

1.2. Reference Entangled State

The principal input state considered in this work is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle), \quad (1.6)$$

or equivalently,

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|HV\rangle + |VH\rangle). \quad (1.7)$$

In the ordered basis of Eq. (1.4), its vector representation is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \quad (1.8)$$

For completeness, the four Bell states are

$$|\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}, \quad |\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}. \quad (1.9)$$

The Bell states are examples of bipartite entangled states. A bipartite pure state is separable if it can be written as a product of independent subsystem states,

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle. \quad (1.10)$$

If such a factorization is not possible, the state is entangled [10]. Entanglement therefore describes correlations belonging to the joint quantum state that cannot be reduced to independent descriptions of the two subsystems.

The state $|\Psi^+\rangle$ is maximally entangled and therefore provides both a physically relevant resource and a convenient analytical benchmark for studying fiber-induced transformations. In this case, the relevant information is not contained in the state of either photon individually, but in the correlations shared between the two spatially separated subsystems.

This makes $|\Psi^+\rangle$ particularly suitable for investigating fiber propagation: coherent local transformations may modify the observed correlation pattern without changing the amount of entanglement, whereas decoherence mechanisms can progressively reduce the shared correlations and eventually degrade the distributed entanglement. Monitoring the evolution of the state, initially $|\Psi^+\rangle$, therefore provides a direct way to distinguish reversible polarization transformations from genuine loss of entanglement quality in the fiber link.

Although the theoretical and numerical frameworks apply to arbitrary two-qubit states, $|\Psi^+\rangle$ is used as the reference state throughout most of the work.

1.3. Density-Operator Description

A pure quantum state can be represented either by its state vector $|\psi\rangle$ or by the corresponding density operator

$$\rho = |\psi\rangle\langle\psi|. \quad (1.11)$$

For a bipartite system, the composite state is described by the density operator ρ_{AB} . If the global bipartite state is pure, then

$$\rho_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|. \quad (1.12)$$

The density-operator formulation is equivalent to the state-vector description for pure states, but it extends naturally to situations in which the system is mixed.

A general mixed state may be expressed as

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \quad p_k \geq 0, \quad \sum_k p_k = 1. \quad (1.13)$$

Such states arise naturally when different physical realizations are averaged statistically or when degrees of freedom correlated with the system are not resolved experimentally. Considerably, a density operator does not in general possess a unique ensemble decomposition; different statistical ensembles may represent the same physical state [10].

The density-operator formalism is adopted as the common representation throughout the work and the simulator.

1.3.1. Expectation Values

For a pure state, the expectation value of an observable O is

$$\langle O \rangle = \langle \psi | O | \psi \rangle. \quad (1.14)$$

In density-operator notation, the corresponding general expression is

$$\langle O \rangle = \text{Tr}(\rho O). \quad (1.15)$$

For $\rho = |\psi\rangle\langle\psi|$, Eq. (1.15) reduces directly to the pure-state expression. The same equation remains valid for mixed states and therefore provides the basis for the evaluation of all observables used in this work.

1.3.2. Reduced Density Operators

For a bipartite state ρ_{AB} , the state accessible locally to either subsystem is obtained through the partial trace,

$$\rho_A = \text{Tr}_B(\rho_{AB}), \quad \rho_B = \text{Tr}_A(\rho_{AB}). \quad (1.16)$$

Mathematically, tracing out subsystem B means summing over a complete basis of $\mathcal{H}_B$, thereby eliminating the degrees of freedom associated with B and producing the unique operator on $\mathcal{H}_A$ that reproduces all local measurement statistics on A .

For example, tracing over subsystem B gives

$$\rho_A = \sum_{j=0}^1 (I_A \otimes \langle j|) \rho_{AB} (I_A \otimes |j\rangle). \quad (1.17)$$

For an observable O_A , characterizing subsystem A

$$\text{Tr} [\rho_{AB} (O_A \otimes I_B)] = \text{Tr}(\rho_A O_A). \quad (1.18)$$

For the Bell state $|\Psi^+\rangle$ of Eq. (1.6), the two reduced density operators are

$$\rho_A = \rho_B = \frac{I_2}{2}. \quad (1.19)$$

Thus, each single-photon state is locally maximally mixed even though the complete two-photon state is pure and maximally entangled. The reduced density operators alone therefore do not determine the nature of the global bipartite state; the distinguishing information is contained in the joint correlations and coherences.

Entangled State vs Incoherent Mixture This distinction can be made explicit by comparing the Bell state $|\Psi^+\rangle$ with the corresponding incoherent mixture having the same computational-basis populations,

$$\rho_{\Psi^+} = |\Psi^+\rangle\langle\Psi^+| = \frac{1}{2} (|01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10|), \quad (1.20)$$

and

$$\rho_{\text{mix}} = \frac{1}{2} (|01\rangle\langle 01| + |10\rangle\langle 10|). \quad (1.21)$$

Both states yield identical reduced density operators,

$$\rho_A = \rho_B = \frac{I_2}{2}, \quad (1.22)$$

but they differ fundamentally at the global level. Their main properties are summarized in Table 1.1.

The comparison shows that identical local density operators, and even

Table 1.1: Comparison between the Bell state $|\Psi^+\rangle$ and the corresponding incoherent mixture with identical computational-basis populations.

Property	ρ_{Ψ^+}	ρ_{mix}
Global state	Pure	Mixed
Reduced states	$\rho_A = \rho_B = I_2/2$	$\rho_A = \rho_B = I_2/2$
Off-diagonal coherence	Present	Absent
Entanglement	Maximal	None
Concurrence	$C = 1$	$C = 0$
Computational-basis populations	$P_{01} = P_{10} = 1/2$	$P_{01} = P_{10} = 1/2$

identical computational-basis populations, do not determine the global quantum state. The distinguishing information is contained in the off-diagonal coherence terms and in the resulting joint correlations.

1.3.3. Bloch Representation

Any single-qubit density operator can be written as

$$\rho = \frac{1}{2} (I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad (1.23)$$

where

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$$

is the vector of Pauli matrices and $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector.

Physical states satisfy

$$\|\mathbf{r}\| \leq 1. \quad (1.24)$$

Pure states satisfy $\|\mathbf{r}\| = 1$ and lie on the surface of the Bloch sphere, whereas mixed states satisfy $\|\mathbf{r}\| < 1$. The maximally mixed state corresponds to

$$\mathbf{r} = 0, \quad \rho = \frac{I_2}{2}.$$

Unitary transformations rotate the Bloch vector while preserving its norm. Decohering processes may reduce its magnitude, with motion toward the interior of the Bloch ball.

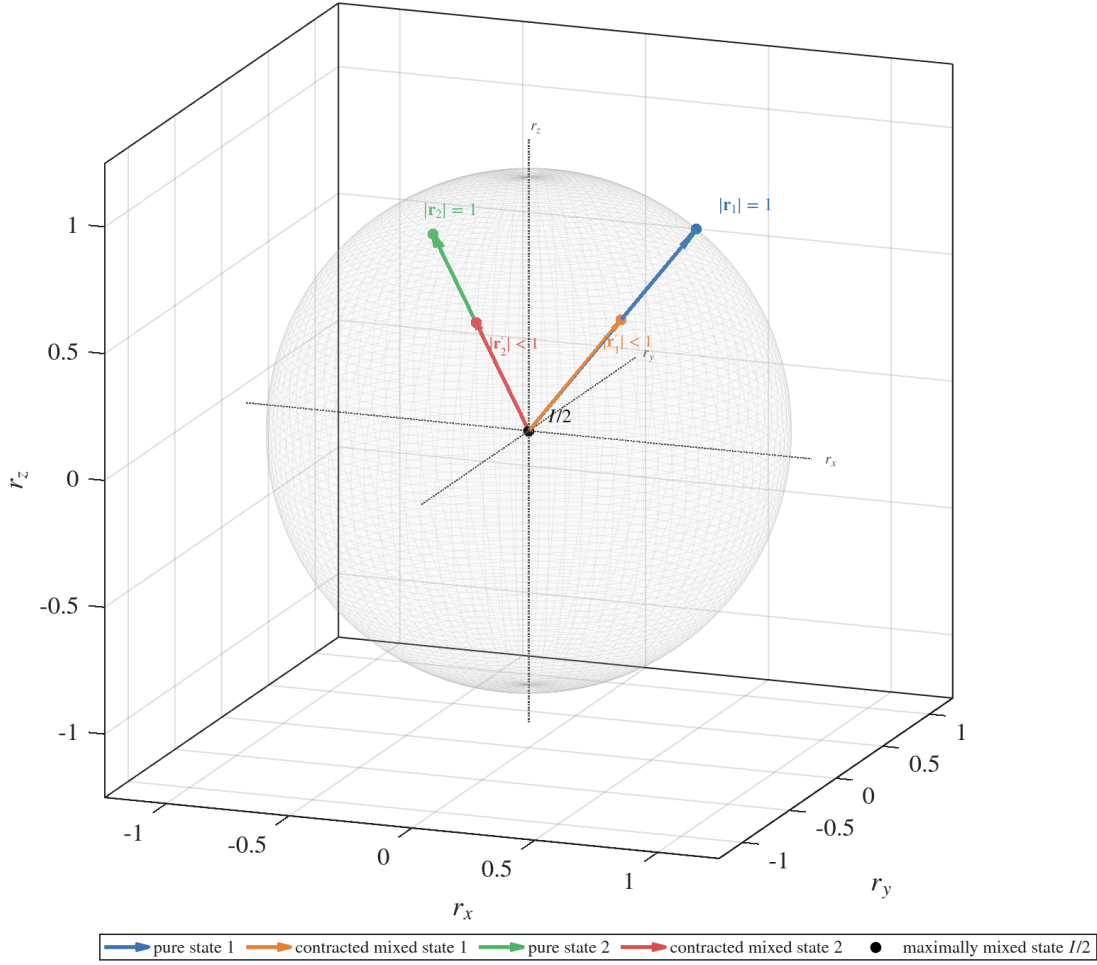


Figure 1.2: Bloch-sphere representation of pure and mixed single-qubit states. Pure states satisfy $\|\mathbf{r}\| = 1$, whereas mixed states are represented by vectors inside the Bloch sphere. The maximally mixed state corresponds to the origin.

1.4. Quantum Evolution and Quantum Channels

The evolution of the two polarization qubits can occur either coherently or through an effective noisy process. These two regimes are naturally distinguished within the density-operator formalism.

1.4.1. Local Unitary Evolution

When each photon undergoes a coherent polarization transformation, its evolution is represented by a unitary operator acting on the corresponding subsystem.

If U_A and U_B denote the transformations acting on arms A and B , respectively, the pure state evolves according to

$$|\psi_{\text{out}}\rangle = (U_A \otimes U_B)|\psi_{\text{in}}\rangle. \quad (1.25)$$

At the density-operator level,

$$\rho_{\text{out}} = (U_A \otimes U_B)\rho_{\text{in}}(U_A^\dagger \otimes U_B^\dagger). \quad (1.26)$$

Local unitary evolution preserves global purity and bipartite entanglement. It may nevertheless modify the correlations observed with respect to fixed laboratory measurement axes. This distinction between a coherent transformation and genuine entanglement degradation is central to the fiber models developed in Chapter 2.

1.4.2. Quantum Channels

A more general transformation of a quantum state is represented by a quantum channel,

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho), \quad (1.27)$$

which is a completely positive and trace-preserving linear map.

A general channel may be written in operator-sum form as [10, 11]

$$\mathcal{E}(\rho) = \sum_{\mu} K_{\mu} \rho K_{\mu}^{\dagger}, \quad \sum_{\mu} K_{\mu}^{\dagger} K_{\mu} = I, \quad (1.28)$$

where the operators K_{μ} are the Kraus operators associated with the map.

Unitary evolution is recovered as the special case containing a single Kraus operator $K = U$. More general channels allow the reduced state to become mixed and are therefore required to describe decoherence, depolarization, and other effective noise mechanisms.

The specific channel models used to represent optical-fiber propagation are introduced in Chapter 2.

1.5. Polarization Correlations and Measurements

For a bipartite polarization state, the physically relevant information is not contained only in the statistics observed separately on arms A and B , but also in how the outcomes of measurements performed on the two photons are related.

A correlation quantifies this statistical relation. If dichotomic polarization measurements are performed on the two arms, with outcomes $a, b \in \{+1, -1\}$, their correlation is defined as the expectation value of the product of the two outcomes. A positive correlation indicates a tendency to obtain equal outcomes, whereas a negative correlation indicates a tendency to obtain opposite outcomes. A vanishing correlation means that the positive and negative contributions balance for the chosen pair of measurement axes.

For measurements described by Pauli operators, these joint statistics are conveniently represented through tensor products of local observables.

1.5.1. Correlation Tensor

For a two-qubit state ρ_{AB} , the correlation tensor is defined by

$$T_{ij} = \text{Tr} [\rho_{AB} (\sigma_i \otimes \sigma_j)], \quad i, j \in \{x, y, z\}. \quad (1.29)$$

Each component T_{ij} is therefore the expectation value of the product of the outcomes obtained when measuring polarization along axis i on photon A and along axis j on photon B . Since the corresponding measurement outcomes are assigned the values ± 1 , one has

$$-1 \leq T_{ij} \leq 1. \quad (1.30)$$

In particular, $T_{ij} = +1$ corresponds to perfect correlation, $T_{ij} = -1$ to perfect anticorrelation, while intermediate values quantify partial correlation for the selected pair of measurement settings.

Equivalently, denoting by $P_{ab}^{(i,j)}$ the joint probability of obtaining outcomes $a, b \in \{+1, -1\}$ when measuring along axes i and j ,

$$T_{ij} = P_{++}^{(i,j)} + P_{--}^{(i,j)} - P_{+-}^{(i,j)} - P_{-+}^{(i,j)}. \quad (1.31)$$

This expression makes explicit that the correlation is determined by joint measurement statistics rather than by the local probabilities associated with either photon separately.

The tensor $T \in \mathbb{R}^{3 \times 3}$ therefore collects the nine bipartite Pauli correlations. A single component $T_{ij} = 0$ does not in general imply that the two subsystems are statistically independent, since correlations may still be present along other pairs of measurement

directions.

Introducing $\sigma_0 = I$, a general two-qubit state can be expanded as

$$\rho_{AB} = \frac{1}{4} \sum_{\mu, \nu \in \{0, x, y, z\}} S_{\mu\nu} \sigma_\mu \otimes \sigma_\nu, \quad (1.32)$$

where

$$S_{\mu\nu} = \text{Tr} [\rho_{AB} (\sigma_\mu \otimes \sigma_\nu)]. \quad (1.33)$$

The quantities $S_{\mu\nu}$ correspond to generalized Stokes parameters. The terms S_{i0} and S_{0j} describe the local Bloch vectors, whereas the 3×3 block S_{ij} coincides with the correlation tensor T .

This representation is particularly useful in polarization experiments because the density operator can be reconstructed from expectation values of local Pauli measurements, as in standard two-qubit quantum-state tomography [12].

The reconstruction follows from the orthogonality relation

$$\text{Tr} [(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)] = 4 \delta_{ik} \delta_{jl}, \quad i, j, k, l \in \{0, x, y, z\}. \quad (1.34)$$

Experimental interpretation. In polarization-entangled photon experiments, the joint probabilities entering Eq. (1.31) are estimated from coincidence counts between detectors associated with the corresponding outcomes on arms A and B . A coincidence identifies two detections occurring within a prescribed temporal window and is interpreted as originating from a single emission of photon pair. After normalization, the measured coincidence counts provide experimental estimates of the joint probabilities required to evaluate the polarization correlations. The practical implementation of this procedure is discussed in chapter 4, concerning experimental activity.

1.5.2. Arbitrary Local Measurement Axes

A projective qubit measurement along a unit vector $\mathbf{n} \in \mathbb{R}^3$ is represented by the observable

$$\sigma(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z, \quad \|\mathbf{n}\| = 1. \quad (1.35)$$

The normalization condition ensures that the observable has eigenvalues ± 1 .

For measurement directions $\mathbf{a}$ and $\mathbf{b}$ on subsystems A and B , respectively, the two-photon correlation is

$$E(\mathbf{a}, \mathbf{b}) = \text{Tr} [\rho_{AB} (\sigma(\mathbf{a}) \otimes \sigma(\mathbf{b}))]. \quad (1.36)$$

Using Eq. (1.29), this becomes

$$E(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top T \mathbf{b}. \quad (1.37)$$

The correlation tensor therefore determines the expectation value associated with any pair of local Pauli measurement directions.

Experimentally, measurements along arbitrary axes can be realized by applying appropriate local polarization transformations before a fixed polarization analysis stage. This provides the connection between the abstract operator description and the wave-plate and polarization-controller settings used in the laboratory.

1.6. Quantitative Characterization of the State

No single scalar quantity completely characterizes a bipartite quantum state. Different measures are therefore used in this work to distinguish mixedness, proximity to a target state and entanglement.

1.6.1. Purity

The purity of a density operator is defined as

$$\gamma(\rho) = \text{Tr}(\rho^2). \quad (1.38)$$

If ρ acts on a d -dimensional Hilbert space,

$$\frac{1}{d} \leq \gamma(\rho) \leq 1. \quad (1.39)$$

The upper bound corresponds to a pure state, whereas the lower bound is attained by the maximally mixed state I_d/d .

If

$$\rho = \sum_i \lambda_i |i\rangle\langle i| \quad (1.40)$$

is the spectral decomposition of the state, then

$$\gamma(\rho) = \sum_i \lambda_i^2. \quad (1.41)$$

For a single qubit written in Bloch form, as $d = 2$, Eq. (1.23) gives

$$\gamma(\rho) = \frac{1 + \|\mathbf{r}\|^2}{2}, \quad \text{with} \quad \frac{1}{2} \leq \gamma(\rho) \leq 1. \quad (1.42)$$

For the bipartite two-qubit polarization system considered throughout this work, $d = 4$, and therefore

$$\frac{1}{4} \leq \gamma(\rho_{AB}) \leq 1. \quad (1.43)$$

Purity quantifies mixedness but does not identify its physical origin. A mixed reduced state may result, for example, from tracing over an entangled subsystem, whereas a mixed global state may arise from statistical averaging or interaction with unresolved degrees of freedom.

1.6.2. Fidelity

Throughout this work, fidelity is defined using the squared Uhlmann convention [13], for two states ρ and σ it is given by

$$F(\rho, \sigma) = \left[\text{Tr} \left(\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right) \right]^2. \quad (1.44)$$

When the reference state is pure, $\sigma = |\phi\rangle\langle\phi|$, the expression simplifies to

$$F(\rho, |\phi\rangle) = \langle\phi|\rho|\phi\rangle. \quad (1.45)$$

In this thesis the principal reference is $|\Psi^+\rangle$, so that

$$F_{\Psi^+} = \langle \Psi^+ | \rho | \Psi^+ \rangle. \quad (1.46)$$

Fidelity measures similarity to a chosen target and must therefore not be interpreted directly as an entanglement measure. A local unitary transformation can substantially change F_{Ψ^+} while leaving the amount of entanglement unchanged.

1.6.3. Concurrence and Tangle

The qualitative distinction between separable and entangled states was introduced in Section 1.2. A quantitative measure is now required to characterize the degree of entanglement of the propagated two-qubit state. Throughout this work, concurrence is adopted as the principal measure of bipartite entanglement [14].

For a pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \quad (1.47)$$

the concurrence is

$$C(|\psi\rangle) = 2|ad - bc|. \quad (1.48)$$

It satisfies

$$0 \leq C \leq 1, \quad (1.49)$$

with $C = 0$ for separable pure states and $C = 1$ for maximally entangled two-qubit states.

For pure bipartite states, concurrence can also be expressed through either reduced density operator,

$$\begin{aligned} C(|\psi_{AB}\rangle) &= 2\sqrt{\det \rho_A} = \sqrt{2[1 - \text{Tr}(\rho_A^2)]} \\ &= 2\sqrt{\det \rho_B} = \sqrt{2[1 - \text{Tr}(\rho_B^2)]}. \end{aligned} \quad (1.50)$$

The equivalence follows from the fact that, for a globally pure bipartite state, ρ_A and ρ_B have the same nonzero eigenvalues.

For a mixed two-qubit state, concurrence is defined through the Wootters spin-flip construction [14]. Defining

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y) \rho^* (\sigma_y \otimes \sigma_y), \quad (1.51)$$

and denoting by $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$ the square roots of the eigenvalues of $\rho \tilde{\rho}$, one obtains

$$C(\rho) = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4). \quad (1.52)$$

The associated tangle is defined as

$$\tau(\rho) = C(\rho)^2. \quad (1.53)$$

For the reference Bell state,

$$C(|\Psi^+\rangle) = 1, \quad \tau(|\Psi^+\rangle) = 1. \quad (1.54)$$

Both quantities are invariant under local unitary transformations, consistently with the invariance of bipartite entanglement under local changes of polarization basis.

1.6.4. Entanglement of Formation

The entanglement of formation provides a resource-oriented measure of bipartite entanglement. For two qubits, it can be expressed analytically in terms of the concurrence [14] as

$$E_F(\rho) = h\left(\frac{1 + \sqrt{1 - C(\rho)^2}}{2}\right), \quad (1.55)$$

where

$$h(x) = -x \log_2 x - (1 - x) \log_2 (1 - x) \quad (1.56)$$

is the binary entropy.

For two-qubit states,

$$0 \leq E_F \leq 1, \quad (1.57)$$

with $E_F = 0$ for separable states and $E_F = 1$ for maximally entangled Bell states. Because E_F is a monotonic function of concurrence for two-qubit states, the two quantities preserve the same ordering of entanglement, while providing different quantitative representations.

1.7. Complementarity of the State Metrics

The quantities introduced in the previous sections characterize different properties of the same density operator and should therefore not be interpreted interchangeably. Purity quantifies whether a state is pure or mixed but does not measure entanglement, while fidelity quantifies similarity with respect to a selected reference state and can change under coherent local transformations. Concurrence, tangle, and entanglement of formation instead quantify bipartite entanglement.

This distinction is particularly relevant for optical-fiber propagation. A coherent local polarization transformation may alter the correlations measured along fixed laboratory axes and reduce fidelity with respect to the initial Bell state without changing purity or entanglement. By contrast, unresolved statistical fluctuations or coupling between polarization and unobserved degrees of freedom may produce mixedness and a reduction of the distributed entanglement.

The simultaneous evaluation of these quantities therefore provides a state-level description capable of distinguishing coherent transformations from decoherence and entanglement degradation. Their operational significance, however, depends on the application for which the distributed state is intended. For this reason, the following section introduces the BBM92 protocol and the quantities used in this work to connect the characterization of the quantum state with the requirements of entanglement-based quantum communication.

1.8. BBM92 and Operational Requirements

The main operational reference considered in this thesis is the BBM92 entanglement-based quantum key distribution protocol [3]. In its polarization implementation, a source distributes the two photons of an entangled pair to two spatially separated parties, conventionally denoted Alice and Bob. Each party independently selects between alternative polarization measurement bases. Measurements performed using compatible bases provide correlated outcomes that can be used for key generation and for estimating the

disturbance introduced along the communication link.

A conceptual representation of the polarization-encoded BBM92 architecture adopted as operational reference throughout this work is shown in Fig. 1.3.

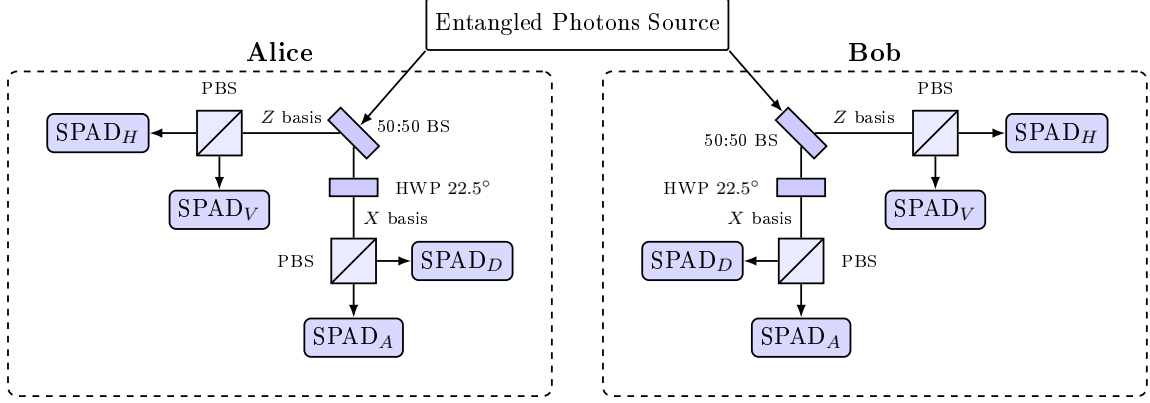


Figure 1.3: Conceptual polarization-encoded BBM92 architecture used as the operational reference in this work. The entangled-photon source distributes one photon through fiber arm A to Alice and the second through fiber arm B to Bob. At each receiver, a 50:50 beam splitter (BS) provides passive basis selection. The direct polarization beam splitter (PBS) analyzes the computational $Z = \{H, V\}$ basis, whereas a half-wave plate (HWP) followed by a PBS enables measurement in the diagonal $X = \{D, A\}$ basis. The resulting polarization components are detected by single-photon avalanche diodes (SPADs).

For the polarization encoding adopted throughout this work, the computational basis has already been identified with

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle,$$

while the complementary diagonal basis is

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \quad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \quad (1.58)$$

For an ideal entangled state, Alice and Bob obtain the expected correlations in both measurement bases. Propagation through the optical link may modify these correlations and introduce errors. Their magnitude is described by the basis-dependent quantum bit error rates

$$Q_Z, \quad Q_X, \quad (1.59)$$

associated with measurements in the computational and diagonal bases, respectively. These quantities can be obtained from the joint measurement probabilities of the propa-

gated two-photon state.

The QBER provides the first operational connection between the quantum state distributed through the fiber and the BBM92 protocol. Degradation or misalignment of the expected polarization correlations increases the measured error rates and therefore increases the amount of information that must be removed during the classical post-processing stages required to obtain a secure key. However, QBER alone does not quantify the rate at which secure key bits can ultimately be generated. A complete protocol-level characterization is instead provided by the Secure Key Rate (SKR), which combines the quality of the measured correlations with the detected coincidence rate and the classical costs associated with error correction and privacy amplification.

In the asymptotic BBM92 description [15], the secure key rate can be expressed in the general form

$$R_{\text{SK}} = q R_{\text{cc}} [1 - f(E_{\text{bit}})H_2(E_{\text{bit}}) - H_2(E_{\text{ph}})], \quad (1.60)$$

where R_{cc} denotes the measured coincidence rate, q is the basis-sifting factor, $f(E_{\text{bit}}) \geq 1$ accounts for the inefficiency of practical error-correction procedures, and

$$H_2(x) = -x \log_2 x - (1 - x) \log_2 (1 - x) \quad (1.61)$$

is the binary entropy. The quantity E_{bit} represents the error rate associated with the basis used for key generation, whereas E_{ph} represents the corresponding phase-error rate inferred from measurements in the complementary basis.

For example, when the Z basis is used for key generation, one may identify

$$E_{\text{bit}} = Q_Z, \quad E_{\text{ph}} = Q_X. \quad (1.62)$$

The roles are exchanged if the X basis is instead used for key generation. In the symmetric case $Q_Z = Q_X = Q$, Eq. (1.60) reduces to

$$R_{\text{SK}} = q R_{\text{cc}} [1 - (1 + f(Q)) H_2(Q)]. \quad (1.63)$$

Equation (1.60) clarifies the distinction between the different modeling levels considered in this thesis. The propagated density operator determines the ideal polarization measurement probabilities and therefore the state-induced contributions to Q_Z and Q_X . The complete experimentally achievable SKR additionally depends on quantities that are not determined by the polarization density operator alone, such as source brightness, optical

transmission losses, detector efficiencies, dark-count rates, timing jitter, the coincidence-window duration, and the adopted classical post-processing scheme. These additional contributions are therefore treated as system-level parameters rather than intrinsic properties of the propagated quantum state.

The objective of this work is consequently not only to quantify degradation of the distributed entanglement, but also to determine which physical configurations of the optical link remain compatible with useful BBM92 operation. For this purpose, the physical configuration is represented by the parameter vector

$$\boldsymbol{\xi} = \{\xi_1, \xi_2, \dots, \xi_m\}, \quad (1.64)$$

which collects the relevant source and fiber parameters. Depending on the model, these may include photon spectral bandwidth, spectral correlation, differential group delay, birefringence orientation, and compensation parameters.

For each physical configuration $\boldsymbol{\xi}$, the propagated state is characterized by a corresponding set of quantities

$$\boldsymbol{\lambda}(\boldsymbol{\xi}) = \{\gamma, F, C, Q_Z, Q_X, \dots\}, \quad (1.65)$$

where γ , F , and C characterize the state in terms of mixedness, similarity to the reference state, and entanglement, while Q_Z and Q_X provide the direct interface with the BBM92 security analysis.

The relation between the physical fiber model, the resulting state, and the protocol-level performance can therefore be summarized as

$$\boxed{\boldsymbol{\xi} \longrightarrow \rho_{AB}(\boldsymbol{\xi}) \longrightarrow \boldsymbol{\lambda}(\boldsymbol{\xi}) \longrightarrow R_{\text{SK}}} \quad (1.66)$$

where the final step additionally depends on the remaining source, detection, and post-processing parameters required by the complete BBM92 system model.

A first operational region can be defined directly from the QBER by requiring the relevant error quantities to remain below an adopted security threshold Q_{th} ,

$$\mathcal{R}_Q = \{\boldsymbol{\xi} : Q_Z(\boldsymbol{\xi}) < Q_{\text{th}}, Q_X(\boldsymbol{\xi}) < Q_{\text{th}}\}. \quad (1.67)$$

Here, Q_{th} is not an arbitrary state-quality threshold, but follows from the adopted security model and post-processing assumptions.

At the complete system level, a more direct definition of useful BBM92 operation is

obtained from the condition that a positive secure key rate can be extracted,

$$\boxed{\mathcal{R}_{\text{BBM92}} = \{\boldsymbol{\xi} : R_{\text{SK}}(\boldsymbol{\xi}) > 0\}}. \quad (1.68)$$

More restrictive requirements may be introduced when a minimum useful key throughput $R_{\text{SK,th}} > 0$ is required for a specific application.

Concurrence plays a complementary role in this analysis. It provides a compact measure of the surviving bipartite entanglement and is therefore used extensively to characterize the effect of fiber propagation. However, concurrence, QBER, and SKR quantify different properties and are not generally equivalent. In particular, a coherent polarization rotation may change the QBER measured in fixed bases without changing the amount of entanglement, while two systems producing the same quantum state and the same QBER may yield very different secure key rates because of different optical losses, detector efficiencies, dark counts, or coincidence rates.

For this reason, a universal concurrence threshold for BBM92 operation is not assumed. Instead, the numerical analysis investigates how concurrence and the basis-dependent QBERs vary together for the specific families of states produced by the considered fiber models. When a clear correspondence exists for a particular model, an operational QBER boundary can then be associated with a corresponding value of concurrence. The subsequent inclusion of the remaining experimental parameters allows the same framework to be extended from state-quality maps to SKR-based communication-performance maps.

The resulting hierarchy has a direct practical interpretation: $\boldsymbol{\xi}$ describes the physical source and fiber configuration, ρ_{AB} describes the resulting distributed quantum state, $\boldsymbol{\lambda}$ contains its state-quality and QBER diagnostics, and R_{SK} quantifies the final secure-key throughput once the complete source, link, detector, and post-processing model is specified. This perspective is used in Chapter 3 to construct operational maps and investigate how the boundaries of useful entanglement distribution depend on the physical parameters of the optical link.

2 | Fiber-Channel Modeling

The mathematical framework introduced in Chapter 1 provides the tools required to describe the evolution and characterization of bipartite polarization states. The purpose of the present chapter is to apply these tools to the progressive construction of models for the propagation of polarization-entangled photon pairs through optical fibers.

The modeling strategy proceeds from analytically controlled effective descriptions toward a more physically accurate frequency-dependent representation of fiber propagation. The first models retain only the polarization degree of freedom and separately describe coherent phase evolution, statistical phase fluctuations, and depolarization. These mechanisms are then combined into an effective polarization channel for the two-arm fiber configuration.

The monochromatic approximation is subsequently relaxed by introducing the spectral structure of the photons and the frequency dependence of birefringent fiber propagation. This leads to a polarization-mode-dispersion (PMD) model in which the complete polarization–frequency evolution remains unitary for an ideal lossless fiber, while an effectively mixed polarization state can emerge after the unresolved spectral degree of freedom is traced out.

Figure 2.1 summarizes the modeling hierarchy adopted throughout the chapter. The different levels represent increasing modeling scope rather than a strict physical derivation from one model to the next.

2.1. Residual Phase Model

A general lossless optical fiber may produce an arbitrary unitary transformation of the polarization state. The first propagation model considered in this work is a reduced description of this general evolution and is motivated by the polarization-compensation procedure employed in the experimental setup.

Polarization compensation is used to realign the dominant output polarization basis with the reference basis of the measurement system. Such compensation does not, however, im-

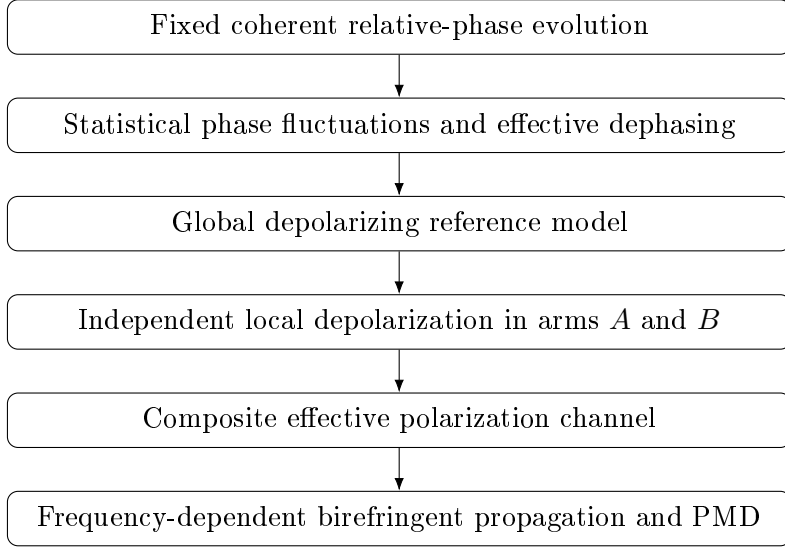


Figure 2.1: Modeling hierarchy adopted in this work. The first levels provide effective polarization-only descriptions of the relevant propagation mechanisms, while the final level explicitly includes the spectral dependence of birefringent fiber propagation.

ply a complete reconstruction or inversion of the full fiber Jones operator¹. In particular, a residual rotation around the stabilized polarization axis may remain.

A limitation of this compensation procedure arises when the polarization state is monitored primarily through measured populations. Since these measurements provide information on the polarization populations but not directly on their relative phase, a residual phase $e^{i\theta}$ may remain unresolved.

A more complete compensation may be achieved by extending the monitoring system to additional polarization bases, since phase-sensitive measurements can provide information on residual transformations that cannot be inferred from computational-basis populations alone.

Within the computational basis adopted in this thesis, the stabilized polarization axis is identified with the z -axis of the Bloch sphere. The remaining coherent transformation can therefore be represented by a relative phase between the horizontal and vertical polarization components, reducing the general polarization transformation to a single phase parameter.

For one polarization qubit, the corresponding unitary operator is

¹The Jones matrix is the 2×2 complex matrix describing the transformation of the polarization amplitudes through an optical element or fiber.

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \quad (2.1)$$

defined up to an irrelevant global phase. Equivalently, this transformation corresponds, up to a global phase factor, to a rotation around the z -axis of the Bloch sphere.

At this modeling level, the phase is assumed to be fixed for each propagation realization. Statistical fluctuations of the same residual phase parameter are introduced separately in the following model.

If the two fiber arms introduce residual phases θ_A and θ_B , respectively, their joint action is

$$U_A(\theta_A) \otimes U_B(\theta_B). \quad (2.2)$$

Applying this transformation to the reference Bell state gives

$$|\psi(\theta_A, \theta_B)\rangle = \frac{1}{\sqrt{2}} (e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle). \quad (2.3)$$

Factoring out the global phase $e^{i\theta_B}$ yields

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} (|01\rangle + e^{i\theta}|10\rangle), \quad \theta = \theta_A - \theta_B. \quad (2.4)$$

The two independent local phase parameters can therefore be reduced to the single effective parameter θ . This relative phase does not modify the computational-basis populations, but it affects phase-sensitive polarization correlations.

With the convention adopted throughout this work, the same state may be generated by assigning the complete relative transformation to arm A ,

$$U_\theta = U_A(\theta) \otimes I_B. \quad (2.5)$$

This convention simplifies the analytical and numerical representation without changing any observable prediction. The parameter θ represents the effective differential residual polarization phase between the two arms.

The term *residual phase model* will be used throughout this work for this one-parameter

effective description. Its relation to the more general fixed-frequency birefringent transformation is derived explicitly in Section 2.7.4.

2.1.1. Correlation Structure

For the state in Eq. (2.4), the correlation tensor defined in Chapter 1 is

$$T(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (2.6)$$

The phase therefore rotates the transverse correlation structure in the xy -plane while leaving the longitudinal anticorrelation unchanged:

$$T_{xx} = T_{yy} = \cos \theta, \quad T_{xy} = -\sin \theta, \quad T_{yx} = \sin \theta, \quad T_{zz} = -1. \quad (2.7)$$

The transformation is local and unitary. Consequently,

$$\gamma = 1, \quad C = 1, \quad \tau = 1, \quad E_F = 1 \quad (2.8)$$

for every value of θ .

The fidelity with respect to the original Bell state is instead

$$F_{\Psi+} = \frac{1 + \cos \theta}{2}. \quad (2.9)$$

Thus, a reduction in fixed-reference Bell-state fidelity does not by itself imply entanglement degradation. It may result entirely from a coherent change in the orientation of the state relative to the selected reference basis.

The residual phase model therefore establishes the first important reference case of this thesis: coherent changes in fixed-basis measurements must be distinguished from genuine decoherence and entanglement degradation.

2.2. Statistical Phase Fluctuations and Dephasing

The fixed-phase description assumes that the relative phase is known and remains constant over the relevant observation interval. In practice, birefringence and optical-path

conditions may vary between successive photon-pair realizations. If these fluctuations are not resolved by the measurement system, the observed state must be obtained by averaging over the corresponding phase realizations.

Let p_k denote the probability associated with the phase θ_k . The ensemble density operator is

$$\rho_{AB}^{\text{ens}} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|, \quad \sum_k p_k = 1. \quad (2.10)$$

The phase statistics can be summarized through the complex coherence parameter

$$\mu = \langle e^{i\theta} \rangle = \sum_k p_k e^{i\theta_k} = \langle \cos \theta \rangle + i \langle \sin \theta \rangle. \quad (2.11)$$

For a continuous probability density $P(\theta)$,

$$\mu = \int d\theta P(\theta) e^{i\theta}. \quad (2.12)$$

It is useful to write

$$\mu = |\mu| e^{i\phi_\mu}. \quad (2.13)$$

The magnitude $|\mu|$ quantifies the coherence surviving the statistical averaging, while the argument ϕ_μ represents the residual coherent phase component of the state.

In the computational basis, the resulting state is

$$\rho_{AB}^{\text{ens}} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (2.14)$$

The diagonal populations remain unchanged, whereas the off-diagonal coherence terms are multiplied by μ . The model therefore corresponds to a random-unitary dephasing process together with a possible residual coherent phase rotation when $\arg(\mu) \neq 0$.

2.2.1. Correlations and State Metrics

The correlation tensor becomes

$$T_\mu = \begin{pmatrix} \text{Re}(\mu) & -\text{Im}(\mu) & 0 \\ \text{Im}(\mu) & \text{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (2.15)$$

The magnitude $|\mu|$ therefore determines the surviving transverse correlation amplitude, whereas the argument of μ determines its remaining orientation.

For this family of states,

$$\gamma = \frac{1 + |\mu|^2}{2}, \quad (2.16)$$

$$F_{\Psi+} = \frac{1 + \text{Re}(\mu)}{2}, \quad (2.17)$$

and

$$C = |\mu|, \quad \tau = |\mu|^2. \quad (2.18)$$

The corresponding entanglement of formation is

$$E_F = h \left(\frac{1 + \sqrt{1 - |\mu|^2}}{2} \right), \quad (2.19)$$

where $h(x)$ is the binary entropy introduced in Chapter 1.

The limiting cases provide a direct physical interpretation. For a deterministic phase, $|\mu| = 1$, and the state remains pure and maximally entangled.

For a phase uniformly distributed over $[0, 2\pi]$, $\mu = 0$, and the transverse coherences vanish completely. The resulting state is

$$\rho_{\mu=0} = \frac{1}{2} (|01\rangle\langle 01| + |10\rangle\langle 10|), \quad (2.20)$$

which retains perfect classical anticorrelation in the computational basis but is no longer

entangled.

Intermediate phase distributions produce partially coherent mixed states.

This model represents decoherence at the reduced polarization level, but it does not yet specify the microscopic physical mechanism responsible for the unresolved phase distribution. A frequency-resolved mechanism producing related losses of polarization coherence is introduced later in this chapter.

2.3. Depolarizing Reference Models

Statistical phase averaging suppresses specific coherence terms while preserving the longitudinal population correlation. A complementary analytical reference is provided by depolarizing channels, which describe effective loss of polarization information directly at the density-operator level.

The depolarizing models introduced here are phenomenological. They do not represent a microscopic model of PMD. Their purpose is to provide analytically controlled descriptions of polarization degradation before the physical origin of decoherence is treated explicitly through frequency-dependent propagation.

Two different classes of effective models are considered. Isotropic depolarization contracts the polarization state uniformly along all Bloch-sphere directions, whereas anisotropic depolarization preserves a preferred polarization axis and suppresses only components orthogonal to it. The latter behavior is particularly relevant as an effective reference for polarization-maintaining or PM-like fiber configurations, where a physically distinguished birefringence axis is present.

2.3.1. Global Depolarizing Baseline

For a two-qubit density operator, the global depolarizing channel is defined as

$$\mathcal{D}_p^4(\rho) = (1 - p)\rho + p\frac{I_4}{4}, \quad 0 \leq p \leq 1. \quad (2.21)$$

For a Bell-state input,

$$\rho(p) = (1 - p)|\Psi^+\rangle\langle\Psi^+| + p\frac{I_4}{4}. \quad (2.22)$$

The corresponding purity is

$$\gamma = 1 - \frac{3}{2}p + \frac{3}{4}p^2, \quad (2.23)$$

the Bell-state fidelity is

$$F_{\Psi^+} = 1 - \frac{3}{4}p, \quad (2.24)$$

and the concurrence is

$$C = \max\left(0, 1 - \frac{3}{2}p\right). \quad (2.25)$$

All bipartite Pauli correlations are contracted according to

$$T_{\text{out}} = (1 - p)T_{\text{in}}. \quad (2.26)$$

The global channel is useful as a validation benchmark, but it is not directly aligned with the geometry of the physical system considered in this thesis, where the two photons propagate through separate fibers.

2.3.2. Local Two-Arm Depolarization

A description more closely aligned with the two-arm geometry assigns an independent depolarizing channel to each propagation arm.

The parameter convention adopted throughout this work is defined directly by the Bloch-vector contraction. A single-qubit isotropic depolarizing channel is written as

$$\mathcal{D}_p(\rho) = (1 - p)\rho + p\frac{I_2}{2}, \quad (2.27)$$

so that $p = 0$ corresponds to no depolarization and $p = 1$ to complete depolarization.

Its linear extension to a generic operator X is

$$\mathcal{D}_p(X) = (1 - p)X + p\frac{I_2}{2}\text{Tr}(X). \quad (2.28)$$

For a single-qubit state written in Bloch form,

$$\rho = \frac{1}{2} (I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad (2.29)$$

the channel gives

$$\mathcal{D}_p(\rho) = \frac{1}{2} [I + (1 - p)\mathbf{r} \cdot \boldsymbol{\sigma}]. \quad (2.30)$$

The Bloch vector therefore undergoes the isotropic contraction

$$\mathbf{r} \longmapsto (1 - p)\mathbf{r}. \quad (2.31)$$

For two independent fiber arms,

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) (\rho_{AB}^{\text{in}}). \quad (2.32)$$

Although the effective noise acts locally, the channel acts on the complete bipartite density operator. The entanglement and two-photon correlations are properties of the joint state and cannot be reconstructed from the two reduced states alone.

2.3.3. Anisotropic Depolarization for a PM-Like Fiber

The isotropic model assumes that polarization information is degraded equally along all directions of the Bloch sphere. This assumption is not appropriate when the propagation medium possesses a physically preferred polarization axis.

A polarization-maintaining or PM-like fiber provides a natural example. Its two principal polarization modes define a distinguished axis in polarization space. When propagation couples polarization to an unresolved degree of freedom, such as frequency or arrival time, coherence between the corresponding principal polarization modes may be reduced while polarization information along the principal axis is preserved.

This behavior is described by an anisotropic dephasing channel. In the standard form commonly introduced for a preferred Z axis, the corresponding Pauli channel preserves the z -component of the Bloch vector while contracting the transverse x - and y -components. An equivalent channel around any other axis is obtained through an appropriate unitary change of basis.

For the convention adopted in this work, the preferred PM-like axis is chosen to be the

X axis. The corresponding effective channel is defined as

$$\mathcal{A}_p^{(x)}(\rho) = \left(1 - \frac{p}{2}\right) \rho + \frac{p}{2} \sigma_x \rho \sigma_x, \quad 0 \leq p \leq 1. \quad (2.33)$$

This parametrization is chosen so that p has the same operational convention used for the isotropic channel: $p = 0$ corresponds to no degradation, while $p = 1$ corresponds to complete suppression of the Bloch components orthogonal to the preferred axis.

The map in Eq. (2.33) is a probabilistic Pauli- X channel. Its action on the Pauli basis is

$$\mathcal{A}_p^{(x)}(I) = I, \quad (2.34)$$

$$\mathcal{A}_p^{(x)}(\sigma_x) = \sigma_x, \quad (2.35)$$

and

$$\mathcal{A}_p^{(x)}(\sigma_y) = (1 - p)\sigma_y, \quad \mathcal{A}_p^{(x)}(\sigma_z) = (1 - p)\sigma_z. \quad (2.36)$$

Therefore, for a Bloch vector

$$\mathbf{r} = \begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix}, \quad (2.37)$$

the channel produces

$$\mathbf{r} \mapsto \begin{pmatrix} r_x \\ (1 - p)r_y \\ (1 - p)r_z \end{pmatrix}. \quad (2.38)$$

The Bloch sphere is thus contracted into an ellipsoid whose major axis coincides with the preferred X direction. States lying on this axis are unaffected, whereas components orthogonal to it progressively lose coherence.

This behavior differs fundamentally from isotropic depolarization. In the isotropic case,

$$(r_x, r_y, r_z) \mapsto (1-p)(r_x, r_y, r_z), \quad (2.39)$$

whereas the anisotropic PM-like model gives

$$(r_x, r_y, r_z) \mapsto (r_x, (1-p)r_y, (1-p)r_z). \quad (2.40)$$

The choice of the X axis is a convention rather than a restriction of the model. For a generic unit vector $\mathbf{n}$, the corresponding preferred-axis Pauli operator is

$$\sigma_{\mathbf{n}} = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (2.41)$$

and the anisotropic channel can be written as

$$\mathcal{A}_p^{(\mathbf{n})}(\rho) = \left(1 - \frac{p}{2}\right) \rho + \frac{p}{2} \sigma_{\mathbf{n}} \rho \sigma_{\mathbf{n}}. \quad (2.42)$$

In Bloch-vector form, decomposing

$$\mathbf{r} = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp}, \quad (2.43)$$

with

$$\mathbf{r}_{\parallel} = (\mathbf{r} \cdot \mathbf{n})\mathbf{n}, \quad \mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}, \quad (2.44)$$

the channel becomes

$$\mathbf{r} \mapsto \mathbf{r}_{\parallel} + (1-p)\mathbf{r}_{\perp}. \quad (2.45)$$

This expression makes the physical interpretation explicit: polarization information parallel to the preferred axis survives, whereas coherence in the orthogonal plane is progressively suppressed.

For two independent PM-like fiber arms with the same preferred X axis,

$$\rho_{AB}^{\text{out}} = (\mathcal{A}_{p_A}^{(x)} \otimes \mathcal{A}_{p_B}^{(x)}) (\rho_{AB}^{\text{in}}). \quad (2.46)$$

For the reference Bell state $|\Psi^+\rangle$, whose correlation tensor is

$$T_{\Psi^+} = \text{diag}(1, 1, -1), \quad (2.47)$$

the anisotropic two-arm channel gives

$$T_{\text{out}} = \text{diag}(1, \eta_{\perp}, -\eta_{\perp}), \quad (2.48)$$

with

$$\eta_{\perp} = (1 - p_A)(1 - p_B). \quad (2.49)$$

Consequently,

$$\langle \sigma_x \otimes \sigma_x \rangle_{\text{out}} = 1, \quad (2.50)$$

while

$$\langle \sigma_y \otimes \sigma_y \rangle_{\text{out}} = \eta_{\perp}, \quad \langle \sigma_z \otimes \sigma_z \rangle_{\text{out}} = -\eta_{\perp}. \quad (2.51)$$

The preferred-axis correlation therefore survives completely, while correlations in the orthogonal directions are attenuated.

This effective channel should not be interpreted as the microscopic propagation law of a PM fiber. Its role is instead to provide a polarization-only reference model for the anisotropic decoherence that can emerge when a preferred birefringence axis is present and unresolved spectral or temporal degrees of freedom are traced out. The corresponding microscopic description will be developed later through frequency-dependent birefringent propagation and PMD.

2.3.4. Correlation-Survival Factor

For the isotropic local depolarizing channel, the single-qubit action on the Pauli basis is

$$\mathcal{D}_p(I) = I, \quad \mathcal{D}_p(\sigma_i) = (1 - p)\sigma_i, \quad i \in \{x, y, z\}. \quad (2.52)$$

Consequently, each two-body Pauli correlation is multiplied by the common factor

$$\eta = (1 - p_A)(1 - p_B). \quad (2.53)$$

The correlation tensor therefore satisfies

$$T_{\text{out}} = \eta T_{\text{in}}, \quad (2.54)$$

and, for arbitrary local measurement axes,

$$E_{\text{out}}(\mathbf{a}, \mathbf{b}) = \eta E_{\text{in}}(\mathbf{a}, \mathbf{b}). \quad (2.55)$$

The factor η will be used throughout the isotropic effective fiber-channel models as the survival factor of bipartite polarization correlations.

For anisotropic depolarization, however, no single scalar survival factor applies to all correlation components. Instead, the contraction is axis dependent. For the X -oriented PM-like model introduced in Section 2.3.3, the x -component is preserved while the transverse y - and z -components acquire the factor $\eta_{\perp}$. The isotropic and anisotropic channels must therefore be kept distinct in the subsequent analytical and numerical treatments.

2.4. Composite Effective Fiber Channel

The fixed relative-phase model and the local depolarizing model represent two different classes of propagation effects. The former is coherent and rotates the correlation structure, whereas the latter is non-unitary at the polarization level and suppresses its magnitude.

They are combined through the effective channel

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_{\theta} \rho_{AB}^{\text{in}} U_{\theta}^{\dagger} \right]. \quad (2.56)$$

For isotropic depolarization, the local depolarizing channel is covariant under single-qubit unitary transformations. Consequently, for the phase transformation considered here,

$$(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_{\theta} \rho U_{\theta}^{\dagger} \right] = U_{\theta} [(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B})(\rho)] U_{\theta}^{\dagger}. \quad (2.57)$$

Thus, within this isotropic model, the order of phase evolution and depolarization does

not affect the final state. This property is specific to the adopted channel and does not hold for arbitrary anisotropic noise mechanisms.

2.4.1. Fixed-Phase Composite Model

For the maximally entangled input family considered here, the reduced states of both subsystems remain $I_2/2$. The action of the two local depolarizing channels can therefore be written compactly as

$$\rho_{AB}^{\text{out}} = \eta |\psi(\theta)\rangle\langle\psi(\theta)| + (1 - \eta) \frac{I_4}{4}. \quad (2.58)$$

The corresponding correlation tensor is

$$T_{\text{out}} = \eta \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (2.59)$$

The parameter θ therefore controls the orientation of the transverse correlations, whereas η controls their magnitude.

The corresponding purity is

$$\gamma = \frac{1 + 3\eta^2}{4}, \quad (2.60)$$

while the Bell-state fidelity is

$$F_{\Psi^+} = \frac{1 - \eta}{4} + \eta \frac{1 + \cos \theta}{2}. \quad (2.61)$$

The concurrence is

$$C = \max \left(0, \frac{3\eta - 1}{2} \right). \quad (2.62)$$

2.4.2. Statistical Composite Model

The composite description can be extended by allowing the relative phase to fluctuate between realizations. The ensemble-averaged state is

$$\rho_{AB}^{\text{ens}} = \sum_k p_k (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_{\theta_k} \rho_{AB}^{\text{in}} U_{\theta_k}^\dagger \right]. \quad (2.63)$$

Using the covariance property in Eq. (2.57),

$$\rho_{AB}^{\text{ens}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[\sum_k p_k U_{\theta_k} \rho_{AB}^{\text{in}} U_{\theta_k}^\dagger \right]. \quad (2.64)$$

For the random-phase family of Section 2.2, this reduces to

$$\rho_{AB}^{\text{ens}} = \eta \rho_{AB}^\mu + (1 - \eta) \frac{I_4}{4}, \quad (2.65)$$

where ρ_{AB}^μ denotes the state in Eq. (2.14).

The output correlation tensor is

$$T_{\text{out}}^{\text{ens}} = \eta \begin{pmatrix} \text{Re}(\mu) & -\text{Im}(\mu) & 0 \\ \text{Im}(\mu) & \text{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (2.66)$$

The two mechanisms therefore leave distinguishable signatures. Phase averaging suppresses the transverse coherence through μ , whereas local isotropic depolarization multiplies all bipartite correlations by η .

The purity becomes

$$\gamma = \frac{1 + \eta^2 (1 + 2|\mu|^2)}{4}, \quad (2.67)$$

the fidelity is

$$F_{\Psi^+} = \frac{1 - \eta}{4} + \eta \frac{1 + \text{Re}(\mu)}{2}, \quad (2.68)$$

and the concurrence is

$$C = \max \left(0, \eta |\mu| - \frac{1 - \eta}{2} \right). \quad (2.69)$$

At the level of transverse polarization correlations, the progression of the effective models can therefore be summarized as

$$\boxed{e^{i\theta} \longrightarrow \mu = \langle e^{i\theta} \rangle \longrightarrow \eta\mu} \quad (2.70)$$

corresponding respectively to coherent phase evolution, statistical dephasing, and the additional suppression produced by isotropic depolarization. The effective model is analytically transparent and useful for distinguishing coherent phase evolution, statistical dephasing, and isotropic polarization degradation. Its main limitation is that the parameters μ , p_A , and p_B describe phenomenological behavior rather than being derived directly from the spectral and birefringent properties of the physical fibers.

2.5. Frequency-Dependent Polarization Propagation

The effective models introduced in the previous sections treat each photon exclusively as a polarization qubit. This description is sufficient when the polarization transformation can be considered approximately constant over the photon bandwidth. For finite-bandwidth photons propagating through a birefringent fiber, however, different spectral components may experience different polarization transformations.

The local polarization operator must therefore be generalized to

$$U_K \longrightarrow U_K(\omega_K), \quad K \in \{A, B\}. \quad (2.71)$$

For a resolved pair of frequencies, the corresponding two-arm transformation is

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \quad (2.72)$$

The complete polarization–frequency evolution of an ideal lossless fiber remains unitary. Nevertheless, when $U_K(\omega_K)$ varies over the occupied bandwidth, different spectral components become associated with different polarization transformations. Polarization and frequency can therefore become correlated during propagation.

If the spectral degree of freedom is not resolved by the measurement system, only the reduced polarization state is observed. The resulting polarization evolution may then appear non-unitary even though the complete optical evolution remains unitary.

This provides the physical connection with the effective models introduced previously:

polarization decoherence can emerge from frequency-dependent coherent propagation followed by the neglect of an unresolved spectral degree of freedom.

2.6. Spectral Description of the Photon Pair

To describe frequency-dependent propagation, the Hilbert space associated with each photon must include both spectral and polarization degrees of freedom,

$$\mathcal{H}_K = \mathcal{H}_{\omega_K} \otimes \mathcal{H}_{\text{pol},K}, \quad K \in \{A, B\}. \quad (2.73)$$

The complete two-photon space can consequently be written as

$$\mathcal{H}_{\text{tot}} = (\mathcal{H}_{\omega_A} \otimes \mathcal{H}_{\omega_B}) \otimes (\mathcal{H}_{\text{pol},A} \otimes \mathcal{H}_{\text{pol},B}). \quad (2.74)$$

The finite-bandwidth spectral state of the photon pair is represented as

$$|F\rangle_{\omega_A\omega_B} = \iint d\omega_A d\omega_B F(\omega_A, \omega_B) |\omega_A, \omega_B\rangle, \quad (2.75)$$

where $F(\omega_A, \omega_B)$ is the joint spectral amplitude (JSA). It is normalized according to

$$\iint d\omega_A d\omega_B |F(\omega_A, \omega_B)|^2 = 1. \quad (2.76)$$

The corresponding joint spectral intensity (JSI) is

$$J(\omega_A, \omega_B) = |F(\omega_A, \omega_B)|^2, \quad (2.77)$$

and represents the joint probability density of the two photon frequencies.

For an SPDC source, the joint spectral structure is determined by the pump spectrum and the phase-matching properties of the nonlinear medium. The JSA may be written schematically as

$$F(\omega_A, \omega_B) = \mathcal{N} \alpha_p(\omega_A + \omega_B) \Phi(\omega_A, \omega_B), \quad (2.78)$$

where α_p is the pump spectral envelope, Φ is the phase-matching function, and $\mathcal{N}$ is a

normalization factor [16].

The resulting frequencies may be statistically correlated. In particular, a narrow pump spectrum constrains the sum frequency $\omega_A + \omega_B$, producing the frequency anticorrelation commonly observed in SPDC photon pairs.

2.6.1. Input Polarization–Frequency State

The model developed in this work assumes that the spectral and polarization degrees of freedom are initially separable. In particular, the polarization state is assumed to be the same for every generated frequency pair.

For the reference Bell-state input,

$$|\Psi_{\text{tot}}^{\text{in}}\rangle = |F\rangle_{\omega_A\omega_B} \otimes |\Psi^+\rangle_{\text{pol}}. \quad (2.79)$$

More generally,

$$\rho_{\text{tot}}^{\text{in}} = |F\rangle\langle F| \otimes \rho_{AB}^{\text{pol},\text{in}}. \quad (2.80)$$

This factorization is an explicit modeling assumption. It neglects initial polarization–frequency distinguishability at the source, so that any coupling between these degrees of freedom appearing after transmission is attributed to the frequency dependence of the fiber propagation [5, 7].

2.6.2. Frequency-Dependent Two-Photon Evolution

Each spectral component of the input state undergoes the corresponding local polarization transformations,

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \quad (2.81)$$

After propagation, the complete polarization–frequency state remains unitary. If frequency is not resolved, the observable polarization state is obtained by tracing over the spectral degrees of freedom,

$$\rho_{AB}^{\text{pol},\text{out}} = \text{Tr}_{\omega_A, \omega_B} [\rho_{\text{tot}}^{\text{out}}]. \quad (2.82)$$

Using the factorized input state of Eq. (2.80), the reduced polarization state becomes

$$\rho_{AB}^{\text{pol,out}} = \iint d\omega_A d\omega_B J(\omega_A, \omega_B) U_{AB}(\omega_A, \omega_B) \rho_{AB}^{\text{pol,in}} U_{AB}^\dagger(\omega_A, \omega_B) \quad (2.83)$$

[5, 9].

Equation (2.83) provides the central connection between frequency-dependent fiber propagation and the effective polarization state. The unresolved spectrum produces a continuous average over the polarization transformations experienced by the different frequency pairs.

The resulting reduced evolution is not in general equivalent to the isotropic depolarizing channel introduced in Section 2.3. Its form depends on the frequency dependence of the fiber transformation. For fixed birefringence axes, for example, the reduced evolution has the form of dephasing in the corresponding eigenpolarization basis.

2.7. Birefringence and Polarization-Mode Dispersion

The frequency dependence of the polarization transformation originates from fiber birefringence. A locally uniform birefringent segment supports two orthogonal polarization eigenmodes, conventionally referred to as slow and fast, with propagation constants $\beta_s(\omega)$ and $\beta_f(\omega)$ [8, 17].

Their differential propagation constant is

$$\Delta\beta(\omega) = \beta_s(\omega) - \beta_f(\omega). \quad (2.84)$$

Using the corresponding effective refractive indices,

$$\beta_{s,f}(\omega) = \frac{\omega}{c} n_{s,f}(\omega), \quad (2.85)$$

one obtains

$$\Delta\beta(\omega) = \frac{\omega}{c} \Delta n_{\text{eff}}(\omega), \quad \Delta n_{\text{eff}} = n_s - n_f. \quad (2.86)$$

For a uniform segment of length L_j , the accumulated differential phase is

$$\Delta\phi_j(\omega) = L_j\Delta\beta_j(\omega). \quad (2.87)$$

Since this phase depends on frequency, the two eigenmodes experience different group delays. The differential group delay (DGD) is determined by

$$\Delta\tau_j = \left. \frac{d\Delta\phi_j}{d\omega} \right|_{\omega_0} = L_j \left. \frac{d\Delta\beta_j}{d\omega} \right|_{\omega_0}. \quad (2.88)$$

Its magnitude $|\Delta\tau_j|$ quantifies the temporal separation accumulated between the two local polarization eigenmodes. The physical relation can therefore be summarized as

$$\boxed{\Delta n_{\text{eff}} \longrightarrow \Delta\beta \longrightarrow \Delta\phi(\omega) \longrightarrow \Delta\tau} \quad (2.89)$$

2.7.1. Single Birefringent Segment

Consider a locally uniform, linear, and lossless birefringent fiber segment. Its polarization evolution can be represented by the frequency-dependent unitary Jones operator

$$U_j(\omega) = e^{i\bar{\phi}_j(\omega)} \exp \left[\frac{i}{2} \Delta\phi_j(\omega) \hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma} \right], \quad (2.90)$$

where $\bar{\phi}_j(\omega)$ is a polarization-independent phase, $\Delta\phi_j(\omega)$ is the differential phase, and $\hat{\mathbf{n}}_j$ identifies the birefringence axis on the Bloch sphere.

Around the carrier frequency ω_0 , the differential phase can be expanded to first order as

$$\Delta\phi_j(\omega) \simeq \Delta\phi_j(\omega_0) + \Delta\tau_j(\omega - \omega_0). \quad (2.91)$$

The first term describes the coherent birefringent phase at the carrier frequency, whereas the second describes its first-order frequency dependence. The latter is responsible for first-order PMD.

At this level, $\hat{\mathbf{n}}_j$ defines the two orthogonal principal polarization directions of the segment, while $|\Delta\tau_j|$ gives their differential group delay.

2.7.2. Fixed-Frequency Limit and Connection to the Residual Phase Model

At a fixed frequency, the transformation in Eq. (2.90) is a coherent unitary polarization transformation. In the local birefringence eigenbasis it can be written, apart from a global phase, as

$$U_j^{\text{eig}}(\omega_0) \sim \begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix}. \quad (2.92)$$

If the relevant eigenbasis is aligned with the computational polarization basis, factoring out the phase of the first component gives

$$U_j^{\text{eig}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta\phi_j} \end{pmatrix}. \quad (2.93)$$

This is the same operator form introduced in the residual phase model of Section 2.1.

For a complete fiber, the local birefringence axes need not coincide with the laboratory H/V basis and may vary along the propagation direction. The fixed-frequency input-output transformation is then obtained from the product of the local segment transformations. After polarization compensation, the reduced model introduced earlier represents the remaining coherent transformation as

$$U_{\text{fib}}^{\text{comp}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \quad (2.94)$$

where θ is the residual coherent phase not removed by the compensation procedure.

The residual phase model is therefore an effective fixed-frequency description of the compensated fiber transformation. At fixed frequency this evolution remains unitary and cannot by itself reduce polarization purity or entanglement. Decoherence emerges when the frequency dependence of the transformation becomes relevant over the finite photon bandwidth and the unresolved spectral degree of freedom is traced out.

2.7.3. First-Order PMD of a Single Segment

It is convenient to introduce the angular-frequency offset from the carrier,

$$\Omega = \omega - \omega_0. \quad (2.95)$$

Using the first-order expansion of Eq. (2.91), the polarization-dependent part of the segment transformation becomes

$$\mathcal{U}_j(\Omega) \simeq \exp \left[\frac{i}{2} (\Delta\phi_j(\omega_0) + \Omega\Delta\tau_j) \hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma} \right]. \quad (2.96)$$

If the polarization transformation at the carrier frequency is compensated, the remaining first-order frequency-dependent transformation is

$$\boxed{\mathcal{U}_j^{\text{res}}(\Omega) = \exp \left[\frac{i}{2} \Omega \Delta\tau_j \hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma} \right]}. \quad (2.97)$$

Let $|s_+\rangle$ and $|s_-\rangle$ denote the eigenstates of the local birefringence generator,

$$(\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}) |s_{\pm}\rangle = \pm |s_{\pm}\rangle. \quad (2.98)$$

An arbitrary input polarization can be written as

$$|\chi\rangle = \alpha |s_+\rangle + \beta |s_-\rangle. \quad (2.99)$$

For a finite-bandwidth photon with spectral amplitude $f_\Omega(\Omega)$, the propagated state is

$$|\Psi_{\text{out}}\rangle = \int d\Omega f_\Omega(\Omega) |\omega_0 + \Omega\rangle \otimes [\alpha e^{+i\Omega\Delta\tau_j/2} |s_+\rangle + \beta e^{-i\Omega\Delta\tau_j/2} |s_-\rangle]. \quad (2.100)$$

The two eigenpolarization components therefore acquire opposite frequency-dependent phases. In the time domain, this corresponds to two wave-packet components separated by the differential group delay $|\Delta\tau_j|$, as illustrated in Fig. 2.2.

If frequency is not resolved, tracing over the spectral degree of freedom gives

$$\rho_{\text{pol}}^{\text{out}} = \text{Tr}_\omega [|\Psi_{\text{out}}\rangle\langle\Psi_{\text{out}}|]. \quad (2.101)$$

In the local eigenpolarization basis,

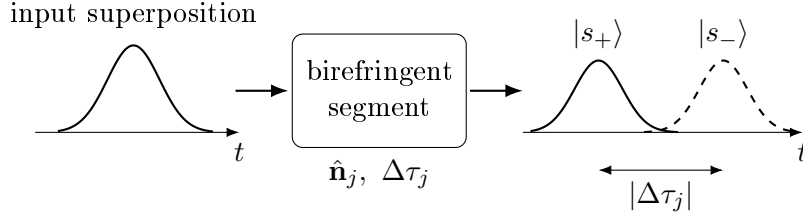


Figure 2.2: Time-domain interpretation of first-order PMD in a single birefringent segment. The two local eigenpolarization components are separated by the differential group delay $|\Delta\tau_j|$.

$$\rho_{\text{pol}}^{\text{out}} = \begin{pmatrix} \rho_{++}^{\text{in}} & \kappa_j \rho_{+-}^{\text{in}} \\ \kappa_j^* \rho_{-+}^{\text{in}} & \rho_{--}^{\text{in}} \end{pmatrix}, \quad (2.102)$$

where

$$\kappa_j = \int d\Omega |f_\Omega(\Omega)|^2 e^{i\Omega\Delta\tau_j}. \quad (2.103)$$

The magnitude $|\kappa_j|$ therefore quantifies the polarization coherence remaining after the unresolved spectral degree of freedom has been traced out [5].

For a Gaussian spectral probability density,

$$|f_\Omega(\Omega)|^2 = \frac{1}{\sqrt{2\pi}\sigma_\omega} \exp\left[-\frac{\Omega^2}{2\sigma_\omega^2}\right], \quad (2.104)$$

the coherence integral in Eq. (2.103) is the characteristic function of the Gaussian spectral distribution and gives

$$|\kappa_j| = \exp\left[-\frac{1}{2}\sigma_\omega^2\Delta\tau_j^2\right]. \quad (2.105)$$

This Gaussian decay of polarization coherence with increasing spectral width and differential group delay is the analytical single-segment limit of the frequency-dependent PMD model considered here [5, 7].

It is convenient to introduce the dimensionless normalized PMD strength

$$\boxed{\chi_j = \sigma_\omega |\Delta\tau_j|}. \quad (2.106)$$

The coherence factor can then be written simply as

$$\boxed{|\kappa_j| = e^{-\chi_j^2/2}}. \quad (2.107)$$

The parameter χ_j compares the PMD-induced differential delay with the spectral scale of the photon. The same DGD therefore produces negligible polarization decoherence for a sufficiently narrow spectrum and increasingly strong decoherence as the occupied bandwidth increases.

For the single-arm configuration considered later in the numerical analysis, with the reference Bell state $|\Psi^+\rangle$ and one photon undergoing the fixed-axis PMD transformation while the other arm remains ideal, the spectral trace produces the same dephased Bell-state family introduced in Section 2.2, with

$$|\mu| = |\kappa| = e^{-\chi^2/2}. \quad (2.108)$$

Using Eq. (2.18), the concurrence is therefore

$$\boxed{C(\chi) = e^{-\chi^2/2}}. \quad (2.109)$$

This result provides the analytical reference for the single-arm PMD numerical experiment of Chapter 3.

2.7.4. Fixed-Frequency Limit and Connection to the Residual Phase Model

At a fixed frequency, the birefringent transformation remains fully unitary. In the local eigenpolarization basis it can be written, apart from a global phase, as

$$U_j^{\text{eig}}(\omega_0) \sim \begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix}. \quad (2.110)$$

When the relevant basis is aligned with the computational polarization basis, this reduces to

$$U_j^{\text{eig}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta\phi_j} \end{pmatrix}. \quad (2.111)$$

This is the same operator form introduced in the residual phase model of Section 2.1. More generally, polarization compensation acts on the complete fixed-frequency fiber transformation and leaves, within the reduced description adopted in this work,

$$U_{\text{fib}}^{\text{comp}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \quad (2.112)$$

The residual phase model can therefore be interpreted as an effective fixed-frequency limit of the compensated fiber transformation. The additional frequency dependence introduced by PMD is what allows polarization coherence to decrease after spectral averaging.

2.8. Concatenated Birefringent Fiber

The birefringence of a real fiber generally varies along the propagation direction. A nonuniform fiber can therefore be represented as a sequence of locally uniform birefringent segments, each characterized by its own birefringence axis and differential phase [8, 17].

Figure 2.3 illustrates this representation.

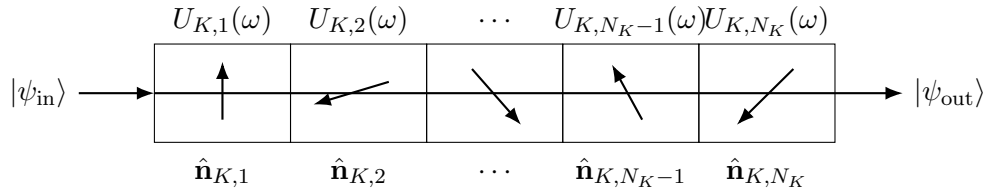


Figure 2.3: Segmented representation of a nonuniform birefringent fiber. Each section is characterized by its own birefringence axis and frequency-dependent unitary transformation. The complete fiber transformation is obtained from the ordered product of the segment operators.

For arm K , let N_K denote the number of segments. The complete frequency-dependent transformation is

$$U_K(\omega_K) = U_{K,N_K}(\omega_K) \cdots U_{K,2}(\omega_K) U_{K,1}(\omega_K). \quad (2.113)$$

The ordering is essential because segments with different birefringence axes do not gen-

erally commute,

$$U_{K,j}(\omega)U_{K,k}(\omega) \neq U_{K,k}(\omega)U_{K,j}(\omega). \quad (2.114)$$

A particular fiber realization is therefore specified by the set of local birefringence axes, differential phases, and differential delays of its segments. For a fixed realization and resolved frequency, the complete propagation remains unitary.

Statistical sampling of the segment parameters can instead be used to represent different possible spatial birefringence profiles. This statistical description of the fiber must be distinguished from the reduced polarization mixedness produced by tracing over unresolved frequency.

2.9. PMD Acting on Polarization-Entangled Photon Pairs

For the two-arm system, each resolved frequency pair undergoes the local transformation

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \quad (2.115)$$

After propagation and spectral tracing, the observable polarization state is

$$\rho_{AB}^{\text{pol,out}} = \iint d\omega_A d\omega_B J(\omega_A, \omega_B) [U_A(\omega_A) \otimes U_B(\omega_B)] \rho_{AB}^{\text{pol,in}} [U_A(\omega_A) \otimes U_B(\omega_B)]^\dagger.$$

(2.116)

For every individual frequency pair, the transformation remains a product of local unitary operators. The reduced polarization state can nevertheless become mixed because the unresolved spectral components undergo different local transformations before being combined by the spectral trace [5, 7, 9].

The joint spectral distribution also determines how the two local frequency-dependent transformations are sampled. If

$$J(\omega_A, \omega_B) = J_A(\omega_A)J_B(\omega_B), \quad (2.117)$$

the effective polarization channel factorizes into independently averaged local channels,

$$\mathcal{E}_{AB} = \mathcal{E}_A \otimes \mathcal{E}_B. \quad (2.118)$$

For spectrally correlated photon pairs, this factorization does not generally hold. Although the physical transformations remain local, the local fiber responses are averaged according to the common joint spectral distribution of the photon pair.

The resulting polarization degradation therefore depends jointly on the source spectrum and on the frequency-dependent transformations of the two fiber arms.

2.10. Relation Between the Modeling Levels

The models developed in this chapter retain progressively different amounts of physical information. The residual phase model isolates coherent polarization evolution, while statistical phase averaging introduces effective dephasing through the coherence parameter μ . The depolarizing models provide analytical references for isotropic polarization degradation, and the composite model combines these effects within a polarization-only description.

The frequency-dependent model instead retains the spectral degree of freedom. Fiber birefringence produces frequency-dependent polarization transformations, while PMD characterizes their first-order variation through the PSPs and the differential group delay. For a fixed lossless fiber realization, the complete polarization–frequency evolution remains unitary, and reduced polarization decoherence emerges after the unresolved spectral degree of freedom is traced out.

The resulting hierarchy can be summarized as

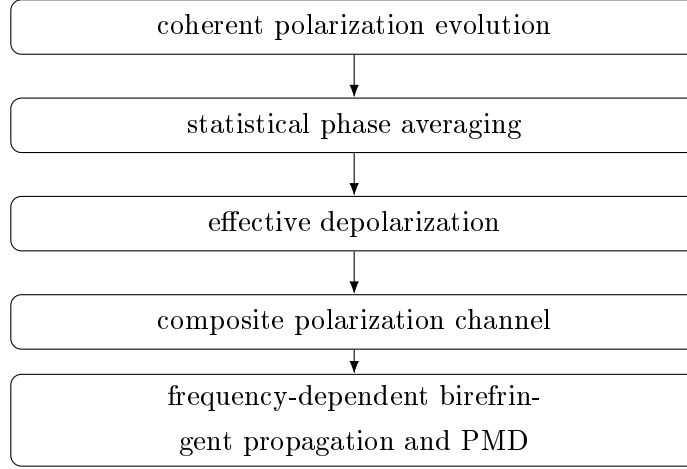


Figure 2.4: Summary of the modeling hierarchy developed in this chapter, from polarization-only effective descriptions to frequency-dependent birefringent propagation and PMD.

The PMD model finally provides the connection between the physical properties of the source and fiber and the reduced polarization state. The relevant physical parameters can therefore be collected in the parameter vector $\boldsymbol{\xi}$, introduced in Section 1.8, while the resulting state and operational quantities are represented by $\boldsymbol{\lambda}(\boldsymbol{\xi})$.

This establishes the physical-to-operational mapping

$$\boxed{\boldsymbol{\xi} \longrightarrow \rho_{AB}^{\text{pol,out}}(\boldsymbol{\xi}) \longrightarrow \boldsymbol{\lambda}(\boldsymbol{\xi})} \quad (2.119)$$

that will be investigated numerically in Chapter 3. In particular, the numerical analysis will determine how variations of the source and fiber parameters modify the distributed polarization state and the corresponding region of parameter space compatible with the adopted BBM92 operational criterion.

2.11. BBM92 Operating Regions of the Fiber Models

The framework introduced in Section 1.8 can now be applied to the specific fiber-channel models developed in this chapter. Each model defines a different parameter space and therefore a corresponding BBM92 operating region.

For the reference state $|\Psi^+\rangle$, measurements in the computational basis are ideally anti-correlated,

$$|\Psi^+\rangle = \frac{|H\rangle_A|V\rangle_B + |V\rangle_A|H\rangle_B}{\sqrt{2}},$$

whereas measurements in the diagonal basis are ideally correlated. Indeed,

$$|\Psi^+\rangle = \frac{|D\rangle_A|D\rangle_B - |A\rangle_A|A\rangle_B}{\sqrt{2}}. \quad (2.120)$$

The definition of an erroneous event must therefore follow the expected correlation pattern in each basis.

In the computational basis, an error corresponds to equal outcomes, HH or VV . The corresponding QBER is therefore

$$Q_Z = P(HH) + P(VV). \quad (2.121)$$

The zz correlation is

$$T_{zz} = P(HH) + P(VV) - P(HV) - P(VH). \quad (2.122)$$

Since the four joint probabilities sum to unity,

$$P(HV) + P(VH) = 1 - [P(HH) + P(VV)], \quad (2.123)$$

and consequently

$$T_{zz} = 2Q_Z - 1. \quad (2.124)$$

It follows that

$$\boxed{Q_Z = \frac{1 + T_{zz}}{2}}. \quad (2.125)$$

In the diagonal basis, the ideal outcomes are instead equal, DD or AA . An error therefore corresponds to the opposite outcomes DA or AD ,

$$Q_X = P(DA) + P(AD). \quad (2.126)$$

The corresponding xx correlation is

$$T_{xx} = P(DD) + P(AA) - P(DA) - P(AD). \quad (2.127)$$

Using normalization gives

$$T_{xx} = 1 - 2Q_X, \quad (2.128)$$

and therefore

$$Q_X = \frac{1 - T_{xx}}{2}. \quad (2.129)$$

Thus, for the $|\Psi^+\rangle$ convention adopted throughout this work, the basis-dependent QBERs can be obtained directly from the corresponding diagonal components of the correlation tensor. These relations provide the connection between the polarization-correlation models developed above and the BBM92 operating regions considered below.

2.11.1. Residual Phase Model

For the fixed residual-phase model,

$$T_{zz} = -1, \quad T_{xx} = \cos \theta. \quad (2.130)$$

The corresponding QBERs are therefore

$$Q_Z(\theta) = 0, \quad Q_X(\theta) = \frac{1 - \cos \theta}{2}. \quad (2.131)$$

The residual phase does not affect the computational-basis outcomes, but it can produce errors in the diagonal basis despite the state remaining maximally entangled.

The corresponding operating region is

$$\mathcal{R}_{\text{BBM92}}^{(\theta)} = \left\{ \theta : \frac{1 - \cos \theta}{2} < Q_{\text{th}} \right\}. \quad (2.132)$$

2.11.2. Statistical Phase Model

For the statistical phase model,

$$T_{zz} = -1, \quad T_{xx} = \text{Re}(\mu), \quad (2.133)$$

and consequently

$$Q_Z(\mu) = 0, \quad Q_X(\mu) = \frac{1 - \text{Re}(\mu)}{2}. \quad (2.134)$$

The BBM92 operating region in the complex coherence-parameter space is therefore

$$\boxed{\mathcal{R}_{\text{BBM92}}^{(\mu)} = \left\{ \mu : \frac{1 - \text{Re}(\mu)}{2} < Q_{\text{th}} \right\}.} \quad (2.135)$$

This result also shows that the operational condition depends on the residual coherent phase contained in μ , and not only on its magnitude $|\mu| = C$.

2.11.3. Local Depolarizing Model

For the local two-arm depolarizing model applied to the reference Bell state,

$$T_{zz} = -\eta, \quad T_{xx} = \eta, \quad (2.136)$$

where

$$\eta = (1 - p_A)(1 - p_B). \quad (2.137)$$

The two measurement bases therefore give the same error rate,

$$Q_Z(\eta) = Q_X(\eta) = \frac{1 - \eta}{2}. \quad (2.138)$$

The operating region is consequently

$$\boxed{\mathcal{R}_{\text{BBM92}}^{(\eta)} = \left\{ \eta : \frac{1 - \eta}{2} < Q_{\text{th}} \right\}.} \quad (2.139)$$

Equivalently, in terms of the individual arm parameters,

$$\mathcal{R}_{\text{BBM92}}^{(p_A, p_B)} = \left\{ (p_A, p_B) : \frac{1 - (1 - p_A)(1 - p_B)}{2} < Q_{\text{th}} \right\}. \quad (2.140)$$

2.11.4. Composite Effective Model

For the statistical composite model developed in Section 2.4.2, the relevant tensor components are

$$T_{zz} = -\eta, \quad T_{xx} = \eta \operatorname{Re}(\mu). \quad (2.141)$$

The corresponding error rates are

$$Q_Z(\eta, \mu) = \frac{1 - \eta}{2}, \quad Q_X(\eta, \mu) = \frac{1 - \eta \operatorname{Re}(\mu)}{2}. \quad (2.142)$$

The operating region can therefore be expressed as

$$\boxed{\mathcal{R}_{\text{BBM92}}^{(\eta, \mu)} = \{(\eta, \mu) : Q_Z(\eta, \mu) < Q_{\text{th}}, Q_X(\eta, \mu) < Q_{\text{th}}\}}. \quad (2.143)$$

This region provides a simple analytical example of an operational boundary in a multi-dimensional channel-parameter space.

2.11.5. Frequency-Dependent PMD Model

The PMD model replaces the phenomenological parameters of the previous descriptions with quantities associated with the source spectrum and the physical fiber propagation.

Let

$$\boldsymbol{\xi}_{\text{PMD}} = \{\sigma_\omega, r, \tau_{\text{DGD},A}, \tau_{\text{DGD},B}, \hat{\mathbf{n}}_A, \hat{\mathbf{n}}_B, \dots\} \quad (2.144)$$

denote the relevant set of physical parameters for a particular PMD model. The reduced polarization state is obtained from Eq. (2.116),

$$\boldsymbol{\xi}_{\text{PMD}} \longrightarrow \rho_{AB}^{\text{pol,out}}(\boldsymbol{\xi}_{\text{PMD}}), \quad (2.145)$$

from which Q_Z and Q_X are evaluated.

Unlike the effective models above, a universal closed-form operating boundary is not expected for a general frequency-dependent fiber. The corresponding region is therefore defined as

$$\mathcal{R}_{\text{BBM92}}^{(\text{PMD})} = \{\boldsymbol{\xi}_{\text{PMD}} : Q_Z(\boldsymbol{\xi}_{\text{PMD}}) < Q_{\text{th}}, Q_X(\boldsymbol{\xi}_{\text{PMD}}) < Q_{\text{th}}\}. \quad (2.146)$$

This formulation converts the PMD model into an operational problem in physical parameter space. The numerical simulations of Chapter 3 are used to evaluate these regions and determine how their boundaries depend on the source, fiber, and compensation parameters.

3 | Numerical Simulations and Results

The fiber-channel models developed in the previous chapter are investigated numerically in this chapter with the objective of determining how the physical properties of the source and optical link affect the operational suitability of the distributed polarization-entangled state.

The numerical analysis is organized around the mapping

$$\boxed{\boldsymbol{\xi} \longrightarrow \boldsymbol{\lambda}(\boldsymbol{\xi})}, \quad (3.1)$$

where $\boldsymbol{\xi}$ collects the relevant source, fiber, and compensation parameters, while $\boldsymbol{\lambda}$ contains the quantities used to characterize the resulting two-photon polarization state. In the present analysis, particular attention is given to purity, fidelity, concurrence, and the basis-dependent quantum bit error rates Q_Z and Q_X .

The central objective is to determine the combinations of physical parameters for which the optical link remains compatible with the adopted BBM92 requirement. The numerical simulations therefore focus on the identification and interpretation of the operating region

$$\boxed{\mathcal{R}_{\text{BBM92}} = \{\boldsymbol{\xi} : Q(\boldsymbol{\xi}) < Q_{\text{th}}\}}, \quad (3.2)$$

where Q_{th} denotes the QBER threshold associated with the security assumptions adopted for the analysis.

The study follows the modeling hierarchy introduced in Chapter 2. Effective polarization-only models are first used to isolate the relation between different degradation mechanisms, state-quality metrics, and BBM92 errors. The analysis is then extended to frequency-dependent birefringent propagation, allowing the operating region to be expressed directly in terms of source and fiber parameters such as spectral bandwidth,

differential group delay, spectral correlation, and the relative properties of the two propagation arms.

Particular attention is devoted to the distinction between state-quality and operational quantities. Concurrence characterizes the surviving bipartite entanglement, while purity and fidelity provide complementary information on the propagated state. BBM92 operation, however, depends directly on the correlations observed in the protocol measurement bases and is therefore assessed through the corresponding QBER. The numerical analysis consequently investigates both the relation and the differences between these descriptions.

The experiments progressively examine the effects of individual channel mechanisms, source–fiber parameter combinations, two-arm propagation, compensation errors, and modeling approximations. The final objective is to construct operating maps that identify the physical configurations belonging to $\mathcal{R}_{\text{BBM92}}$ and quantify how its boundary changes with the properties and modeling of the optical link.

3.1. Computational Framework and Numerical Methodology

All numerical models presented in this chapter are implemented in MATLAB using a modular library that reproduces the mathematical structure introduced in the theoretical framework. Quantum states, density operators, polarization transformations, quantum channels, and measurement observables are represented directly through complex vectors and matrices.

Reusable numerical routines provide the common operations required throughout the analysis, including state preparation, local evolution, quantum-channel application, frequency-dependent fiber propagation, spectral averaging, polarization measurements, and state characterization. The numerical simulations are constructed from these common components, ensuring that different physical configurations and modeling assumptions are evaluated using the same underlying computational framework.

For a specified physical configuration ξ , the numerical workflow is summarized by

$$\boxed{\xi \longrightarrow \rho_{AB}(\xi) \longrightarrow \lambda(\xi) \longrightarrow \mathcal{R}_{\text{BBM92}}.} \quad (3.3)$$

For effective polarization channels, the propagated density operator is obtained directly from the corresponding channel action. For frequency-dependent fiber models, the two-

photon state is propagated over the joint spectrum and the unresolved spectral degree of freedom is subsequently traced out to obtain the reduced polarization density operator.

The resulting state is characterized using a common set of quantities throughout the numerical study. Purity, fidelity, and concurrence describe complementary properties of the distributed quantum state, while the polarization measurement statistics determine Q_Z and Q_X . The latter provide the direct criterion used to classify each simulated physical configuration with respect to the adopted BBM92 operating condition.

The numerical simulations presented below therefore use the simulator primarily as a parameter-space analysis tool. Rather than considering isolated propagation examples, each experiment is designed to determine how a specific physical mechanism or modeling assumption modifies the boundary between operational and non-operational regions of the optical link.

3.2. Effective Fiber Models and BBM92 Performance

3.2.1. Objective and Numerical Setup

The first numerical experiment compares the effective fiber models introduced in Chapter 2 from both a state-characterization and an operational BBM92 perspective. The objective is to distinguish changes in the quality of the distributed entangled state from changes in the correlations observed in the fixed protocol measurement bases.

The input state is $|\Psi+\rangle$ and three mechanisms are considered: coherent relative-phase evolution, statistical dephasing, and symmetric local depolarization. For each propagated state, the purity γ , fidelity F , concurrence C , and basis-dependent error rates Q_Z and Q_X are evaluated.

The operational classification is based on

$$Q_{\text{worst}} = \max(Q_Z, Q_X), \quad Q_{\text{worst}} < Q_{\text{th}}, \quad (3.4)$$

with the threshold adopted in the present numerical analysis set to

$$Q_{\text{th}} = 0.11. \quad (3.5)$$

The coherent-phase model is evaluated over 181 values of $\theta \in [0, \pi]$, while 101 points are used for both the dephasing and depolarization sweeps.

3.2.2. Coherent Relative-Phase Evolution

A local relative phase produces the state

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}.$$

Since this evolution is local and unitary, purity and concurrence remain equal to unity. The fidelity with the initial Bell state and the BBM92 error rates are

$$F = \frac{1 + \cos \theta}{2}, \quad Q_Z = 0, \quad Q_X = \frac{1 - \cos \theta}{2}. \quad (3.6)$$

The numerical results in Figure 3.1 reproduce this behavior. In particular, the state remains maximally entangled while the error rate in the diagonal basis increases with the uncompensated relative phase.

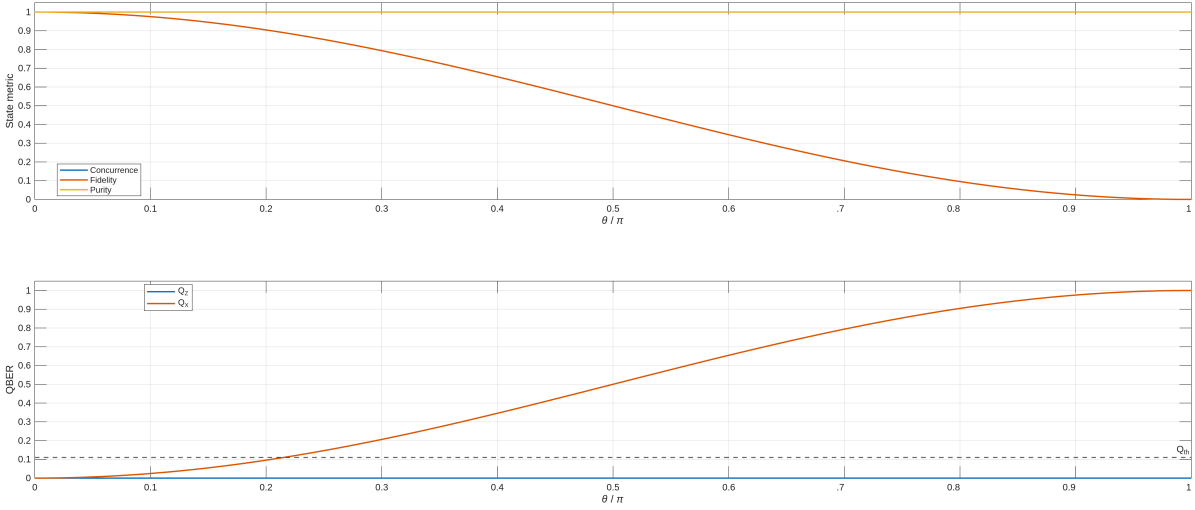


Figure 3.1: Coherent relative-phase evolution. Purity and concurrence remain equal to unity, while fidelity and the diagonal-basis QBER vary with the relative phase. The dashed line indicates the adopted BBM92 QBER threshold.

The corresponding numerical operating boundary is

$$\theta_{th} \simeq 0.6761 \text{ rad} \simeq 0.215\pi. \quad (3.7)$$

This case directly demonstrates that maximal entanglement does not by itself guarantee low QBER when measurements are performed in fixed polarization bases.

3.2.3. Statistical Dephasing

Statistical fluctuations of the relative phase reduce the polarization coherence. For the real coherence parameter $\mu \in [0, 1]$, the resulting state is

$$\rho_\mu = \frac{1}{2} (|01\rangle\langle 01| + |10\rangle\langle 10|) + \frac{\mu}{2} (|01\rangle\langle 10| + |10\rangle\langle 01|). \quad (3.8)$$

For this state family,

$$C = \mu, \quad Q_Z = 0, \quad Q_X = \frac{1 - \mu}{2}. \quad (3.9)$$

As shown in Figure 3.2, decreasing coherence now corresponds to genuine entanglement degradation and increasing error in the diagonal basis.

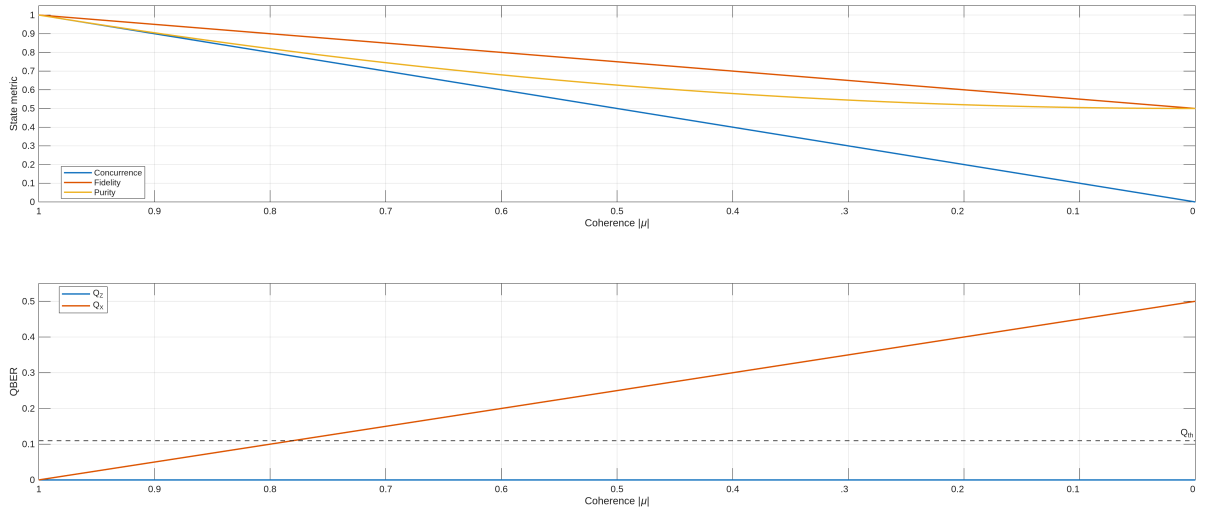


Figure 3.2: Statistical dephasing as a function of the coherence parameter μ . Loss of coherence reduces concurrence, purity, and fidelity while increasing Q_X ; the H/V -basis correlations remain unaffected.

The QBER threshold is crossed at

$$\mu_{th} = 0.780. \quad (3.10)$$

Because $C = \mu$ for this particular family, the same point corresponds to $C = 0.78$. This value is specific to the dephasing model and does not represent a universal concurrence threshold for BBM92 operation.

3.2.4. Symmetric Two-Arm Depolarization

The third model applies the same local depolarization probability to both photons,

$$p_A = p_B = p. \quad (3.11)$$

In this case both measurement bases are affected. As shown in Figure 3.3, increasing p progressively decreases purity, fidelity, and concurrence, while Q_Z and Q_X increase together.

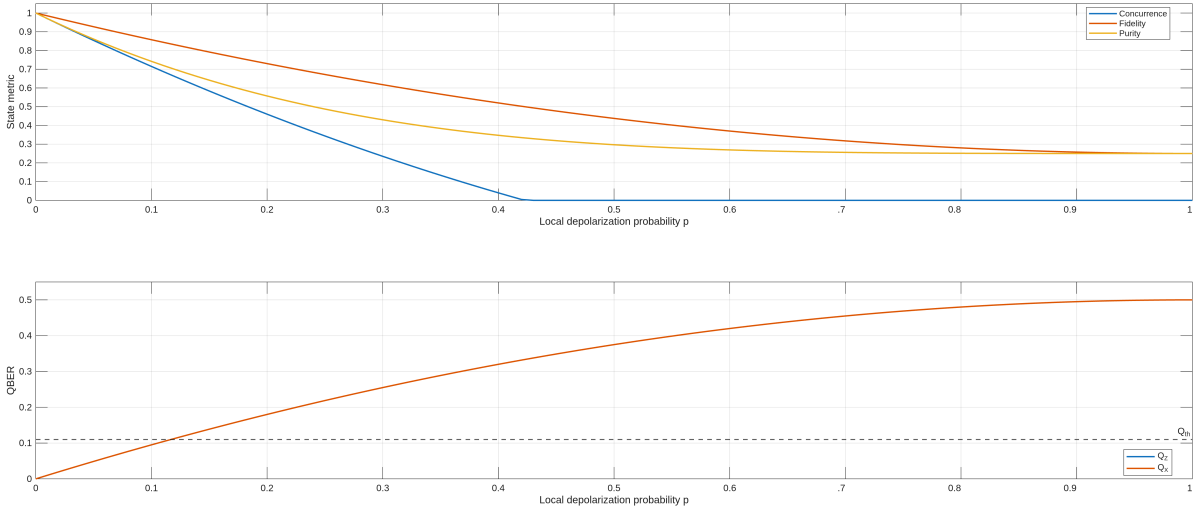


Figure 3.3: Symmetric independent local depolarization. Increasing the local depolarization probability reduces the state-quality metrics and produces equal errors in the two BBM92 measurement bases.

The numerical operating boundary is

$$p_{th} \simeq 0.1168. \quad (3.12)$$

3.2.5. Comparison of State Quality and BBM92 Performance

Figure 3.4 directly compares the worst-basis QBER with concurrence for the three models.

The operating boundaries obtained from the three one-parameter models are

$$\boxed{\theta_{th} \simeq 0.6761 \text{ rad}, \quad \mu_{th} = 0.780, \quad p_{th} \simeq 0.1168.} \quad (3.13)$$

These values parameterize different physical mechanisms and should therefore not be

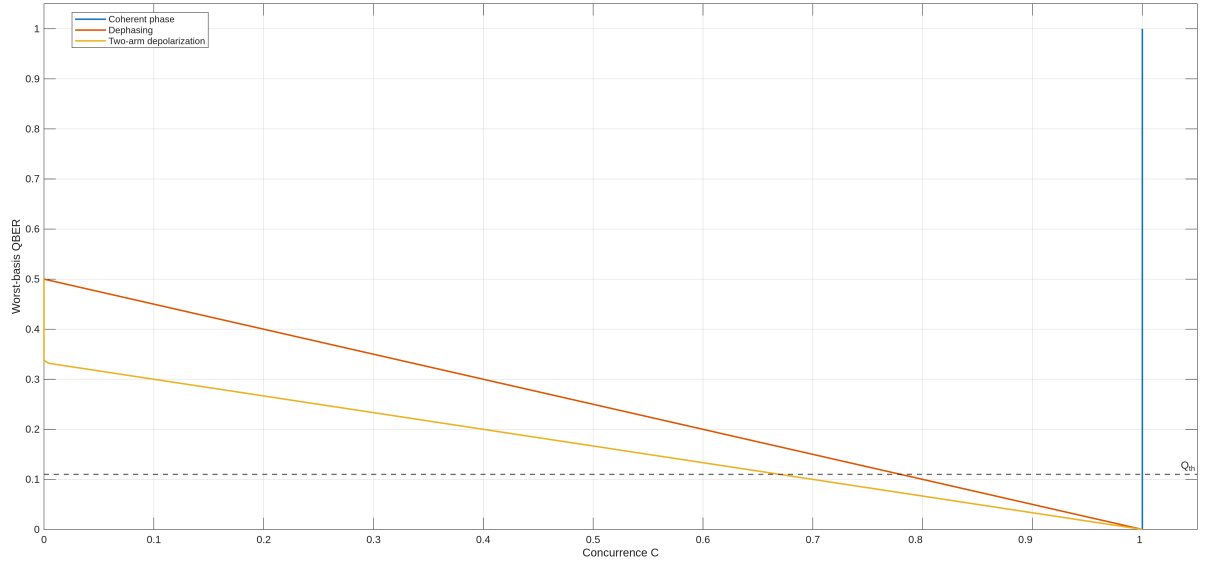


Figure 3.4: Worst-basis QBER as a function of concurrence for the three effective fiber models. The relation between entanglement and BBM92 performance depends on the physical degradation mechanism. Coherent phase evolution provides the limiting example in which $C = 1$ while the QBER varies.

interpreted as equivalent channel thresholds. Their common meaning is that they identify where each model crosses the adopted BBM92 QBER criterion.

The experiment establishes the distinction used throughout the following numerical analysis: purity, fidelity, and concurrence characterize different properties of the distributed quantum state, whereas the BBM92 operating region is determined directly from the measurement error rates. The subsequent experiments extend this criterion from effective channel parameters to the physical source and fiber parameters of the frequency-dependent propagation model.

3.3. Source–Fiber Operating Region under Single-Arm PMD

3.3.1. Objective and Numerical Setup

The second numerical experiment introduces the frequency-dependent PMD model and investigates how the source spectral bandwidth and the fiber differential group delay jointly determine the BBM92 operating region.

A single-arm configuration is considered in order to isolate this dependence. The photon

propagating through arm A experiences first-order PMD with a fixed birefringence axis, while arm B is kept ideal. The polarization transformation at the carrier frequency is assumed to be compensated, so that only the frequency-dependent PMD contribution is retained.

The input polarization state is the reference Bell state $|\Psi^+\rangle$. A Gaussian spectral distribution is used, with spectral bandwidth σ_ω varied over

$$2.5 \times 10^{10} \leq \sigma_\omega \leq 2.0 \times 10^{11} \text{ rad/s}, \quad (3.14)$$

while the differential group delay is varied over

$$0 \leq \Delta\tau \leq 30 \text{ ps}. \quad (3.15)$$

The numerical spectral integration uses 81 frequency nodes per arm over a window of $\pm 6\sigma_\omega$. The PMD axis in arm A is chosen along the x -axis of the Bloch sphere. With this orientation, the diagonal-basis correlation is protected, while the computational-basis correlation is degraded by the frequency-dependent propagation.

For every pair $(\sigma_\omega, \Delta\tau)$, the frequency-resolved polarization state is propagated and the unresolved spectral degree of freedom is traced out according to Eq. (2.116). The resulting polarization state is characterized through concurrence and the two BBM92 error rates Q_Z and Q_X .

The operational classification is based on the worst-basis error

$$Q_{\text{worst}} = \max(Q_Z, Q_X), \quad (3.16)$$

with the adopted threshold

$$Q_{\text{worst}} < Q_{\text{th}}, \quad Q_{\text{th}} = 0.11. \quad (3.17)$$

3.3.2. Source–Fiber BBM92 Operating Region

Figure 3.5 shows the worst-basis QBER over the source–fiber parameter space.

The operating region lies below and to the left of the dashed boundary. For a fixed spectral bandwidth, increasing the DGD increases the frequency dependence of the polarization

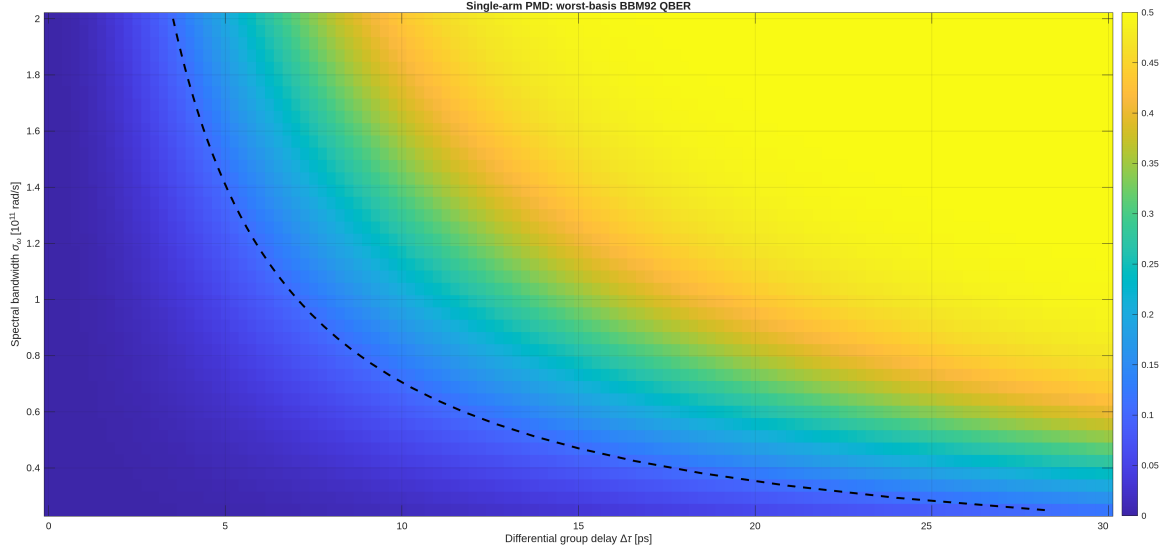


Figure 3.5: Worst-basis BBM92 QBER for single-arm first-order PMD as a function of the photon spectral bandwidth σ_ω and the differential group delay $\Delta\tau$. The dashed curve identifies the adopted BBM92 operating boundary $Q_{\text{worst}} = Q_{\text{th}}$, with $Q_{\text{th}} = 0.11$.

transformation and consequently the QBER. Conversely, for a fixed DGD, a broader photon spectrum samples a larger range of frequency-dependent polarization transformations and therefore produces the same effect.

The boundary consequently exhibits an approximately inverse dependence between the maximum tolerable DGD and the source bandwidth. This behavior follows directly from the normalized PMD parameter introduced in Eq. (2.106),

$$\chi = \sigma_\omega |\Delta\tau|. \quad (3.18)$$

For the Gaussian spectrum considered here, Chapter 2 showed that the surviving polarization coherence is

$$|\kappa| = e^{-\chi^2/2}, \quad (3.19)$$

as given by Eq. (2.107). The numerical operating boundary is therefore expected to correspond to an approximately constant value of χ , rather than to an independent limitation on σ_ω or $\Delta\tau$.

3.3.3. Entanglement at the Operational Boundary

The corresponding concurrence is shown in Fig. 3.6. The same BBM92 operating boundary is superimposed on the state-quality map.

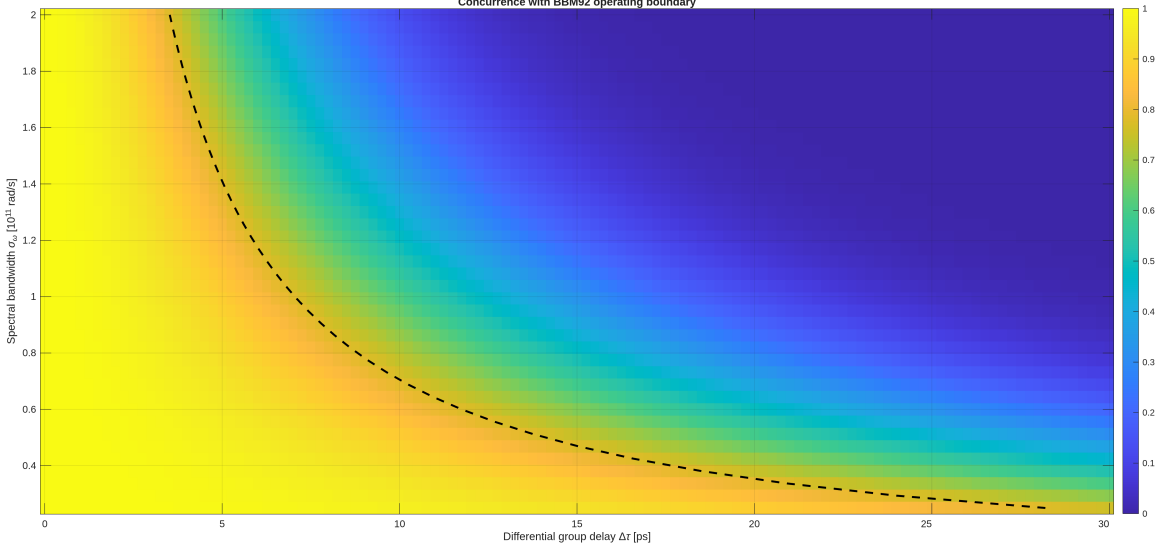


Figure 3.6: Concurrence of the reduced polarization state under single-arm first-order PMD. The dashed curve is the BBM92 operating boundary obtained from $Q_{\text{worst}} = 0.11$, rather than from an imposed concurrence threshold.

The concurrence decreases continuously as the product $\sigma_\omega |\Delta\tau|$ increases. This behavior is consistent with the analytical single-arm Gaussian result derived in Eq. (2.109),

$$C(\chi) = e^{-\chi^2/2}. \quad (3.20)$$

Importantly, the dashed curve in Fig. 3.6 is not defined by selecting an independent minimum acceptable concurrence. It is obtained exclusively from the BBM92 QBER requirement and subsequently represented on the concurrence map. The figure therefore shows the amount of entanglement remaining when the operational QBER boundary is reached.

3.3.4. Operating-Boundary Analysis

The numerical results confirm that the source bandwidth and fiber DGD enter the operational boundary predominantly through the normalized PMD parameter $\chi = \sigma_\omega |\Delta\tau|$.

For the adopted QBER threshold, the analytical single-arm Gaussian model gives

$$\chi_{\text{th}}^{\text{ana}} = 0.704927. \quad (3.21)$$

The numerical parameter sweep gives

$$\bar{\chi}_{\text{th}}^{\text{num}} = 0.704872, \quad \sigma_{\chi_{\text{th}}} = 5.505 \times 10^{-5}, \quad (3.22)$$

with a maximum absolute deviation from the analytical boundary of

$$\max |\chi_{\text{th}}^{\text{num}} - \chi_{\text{th}}^{\text{ana}}| = 2.044 \times 10^{-4}. \quad (3.23)$$

The concurrence evaluated along the numerical BBM92 boundary is correspondingly

$$\bar{C}_{\text{th}} = 0.780000, \quad (3.24)$$

with a standard deviation of only 1.902×10^{-9} over the explored bandwidth range. The approximately constant concurrence along the boundary is a consequence of the single-parameter dependence of this particular fixed-axis Gaussian PMD model and should therefore not be interpreted as a universal concurrence threshold for BBM92 operation.

The simulation also reproduces the expected limiting behavior at zero DGD: the propagated state remains the original Bell state, with unit concurrence and zero QBER in both measurement bases. For the selected x -oriented PMD axis, the diagonal basis remains protected throughout the sweep, $Q_X = 0$, while Q_Z determines the operating boundary.

The experiment therefore reduces the two-dimensional source–fiber parameter space to a physically meaningful normalized condition,

$$\boxed{\sigma_{\omega} |\Delta\tau| < \chi_{\text{th}} \simeq 0.705,} \quad (3.25)$$

for the specific single-arm, fixed-axis, Gaussian PMD configuration considered here. This result provides the first direct source–fiber operating map of the numerical analysis and establishes the reference case for the two-arm and spectrally correlated configurations considered in the following experiment.

3.4. Two-Arm PMD and Spectral-Correlation Operating Regions

3.4.1. Objective and Numerical Setup

The third numerical experiment extends the single-arm analysis of Section 3.3 to the complete two-arm configuration. The objective is to determine how PMD accumulated by both photons combines with the spectral correlations of the entangled-photon source in determining the BBM92 operating region.

The frequency offsets of photons A and B from their respective carrier frequencies are denoted by

$$\Omega_A = \omega_A - \omega_{0,A}, \quad \Omega_B = \omega_B - \omega_{0,B}. \quad (3.26)$$

The joint spectral statistics of the two photons are described by a Gaussian joint spectral intensity of the form

$$J(\Omega_A, \Omega_B) \propto \exp \left[-\frac{(\Omega_A + \Omega_B)^2}{2\sigma_\Sigma^2} - \frac{(\Omega_A - \Omega_B)^2}{2\sigma_\Delta^2} \right], \quad (3.27)$$

where σ_Σ and σ_Δ control the widths along the sum- and difference-frequency directions, respectively.

For equal marginal spectral bandwidths,

$$\sigma_{\omega,A} = \sigma_{\omega,B} = \sigma_\omega, \quad (3.28)$$

the spectral correlation coefficient is defined as

$$r = \frac{\text{Cov}(\Omega_A, \Omega_B)}{\sigma_\omega^2}, \quad -1 < r < 1. \quad (3.29)$$

With the Gaussian convention adopted in the simulator, the corresponding sum- and difference-frequency widths are

$$\sigma_\Sigma = \sigma_\omega \sqrt{2(1+r)}, \quad \sigma_\Delta = \sigma_\omega \sqrt{2(1-r)}. \quad (3.30)$$

The parameter r therefore describes how the frequency offsets of the two photons are statistically related. For $r < 0$, the frequencies are anticorrelated: a positive frequency offset of one photon tends to be accompanied by a negative offset of the other. For $r > 0$, the two offsets tend to have the same sign and are positively correlated. The case $r = 0$ corresponds to statistically uncorrelated frequency offsets.

Equivalently, negative r narrows the distribution along the sum-frequency coordinate $\Omega_A + \Omega_B$, whereas positive r narrows it along the difference-frequency coordinate $\Omega_A - \Omega_B$. The joint spectral distribution therefore determines which combinations of photon frequencies contribute most strongly to the propagated two-photon state.

3.4.2. Generation of the Frequency-Resolved Polarization States

The input polarization state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle),$$

with density operator

$$\rho_{\text{in}} = |\Psi^+\rangle \langle \Psi^+|.$$

For every spectral pair (Ω_A, Ω_B) , the polarization state is propagated through the two frequency-dependent fiber transformations according to

$$\rho_{AB}(\Omega_A, \Omega_B) = U_{AB}(\Omega_A, \Omega_B) \rho_{\text{in}} U_{AB}^\dagger(\Omega_A, \Omega_B), \quad (3.31)$$

where

$$U_{AB}(\Omega_A, \Omega_B) = U_A(\Omega_A) \otimes U_B(\Omega_B). \quad (3.32)$$

Each arm is represented by one first-order PMD segment. Using the fiber model introduced in Chapter 2, the corresponding unitary is

$$U_K(\Omega_K) = \exp \left[\frac{i}{2} \Delta\tau_K \Omega_K (\mathbf{n}_K \cdot \boldsymbol{\sigma}) \right], \quad K \in \{A, B\}, \quad (3.33)$$

where $\Delta\tau_K$ is the differential group delay and $\mathbf{n}_K$ is the local birefringence axis. The

carrier-frequency differential phase is set to zero in the present experiment so that only the frequency-dependent PMD contribution is retained.

Both PMD axes are fixed along the same direction,

$$\mathbf{n}_A = \mathbf{n}_B = \hat{\mathbf{x}}, \quad (3.34)$$

so that the experiment isolates the effects of $\Delta\tau_A$, $\Delta\tau_B$, and the spectral correlation r . Relative PMD-axis misalignment is investigated separately in the following experiment.

The joint spectral distribution in Eq. (3.27) generates a statistical ensemble of frequency-resolved polarization states. Since the photon frequencies are not resolved in the polarization measurement, the reduced polarization density operator is obtained by tracing over the spectral degree of freedom,

$$\rho_{\text{pol}} = \iint d\Omega_A d\Omega_B P(\Omega_A, \Omega_B) \rho_{AB}(\Omega_A, \Omega_B), \quad (3.35)$$

where

$$P(\Omega_A, \Omega_B) = \frac{J(\Omega_A, \Omega_B)}{\iint d\Omega_A d\Omega_B J(\Omega_A, \Omega_B)} \quad (3.36)$$

is the normalized joint spectral probability distribution.

Numerically, the integral in Eq. (3.35) is evaluated over the discrete spectral grid as

$$\rho_{\text{pol}} \simeq \sum_{k,l} Q_{kl} \rho_{AB}(\Omega_{A,k}, \Omega_{B,l}), \quad (3.37)$$

where Q_{kl} are the normalized quadrature-weighted probabilities generated from the Gaussian joint spectral intensity.

This construction makes the physical role of the source spectrum explicit. Different values of r generate different statistical combinations of (Ω_A, Ω_B) , which are subsequently converted by the two fiber transformations into different distributions of polarization transformations. The reduced polarization state therefore depends jointly on the source spectrum and on the PMD parameters of both fiber arms.

3.4.3. Physical Role of Spectral Correlation

For the first-order model of Eq. (3.33), the frequency-dependent differential phases accumulated in the two arms scale as

$$\delta\phi_A = \Delta\tau_A\Omega_A, \quad \delta\phi_B = \Delta\tau_B\Omega_B. \quad (3.38)$$

Consequently, correlations between Ω_A and Ω_B translate directly into correlations between the PMD-induced phase contributions.

For $r < 0$, the two frequency offsets preferentially have opposite signs. When the two DGDs are comparable and the PMD axes are aligned, the corresponding frequency-dependent contributions can therefore partially compensate. For $r > 0$, the frequency offsets preferentially vary in the same direction and the two PMD contributions instead tend to reinforce each other.

The two-arm PMD degradation is consequently determined not only by the individual values of $\Delta\tau_A$ and $\Delta\tau_B$, but also by the joint spectral structure of the photon pair. This is the principal physical difference with respect to the single-arm configuration investigated in the previous experiment.

3.4.4. Numerical Parameter Space

The marginal spectral bandwidth is fixed to

$$\sigma_\omega = 1.0 \times 10^{11} \text{ rad/s}. \quad (3.39)$$

The two differential group delays are independently varied over

$$0 \leq \Delta\tau_A, \Delta\tau_B \leq 20 \text{ ps}, \quad (3.40)$$

using 31 points per arm.

Three representative spectral-correlation conditions are initially considered,

$$r = -0.9, \quad r = 0, \quad r = +0.9, \quad (3.41)$$

corresponding respectively to strongly anticorrelated, uncorrelated, and strongly posi-

tively correlated spectra.

The spectral trace is evaluated using 41 frequency nodes per arm over the interval

$$-6\sigma_\omega \leq \Omega_K \leq 6\sigma_\omega. \quad (3.42)$$

A second numerical sweep varies the spectral correlation continuously over

$$-0.9 \leq r \leq 0.9 \quad (3.43)$$

using 37 points, while restricting the fiber parameters to the symmetric configuration

$$\Delta\tau_A = \Delta\tau_B = \Delta\tau. \quad (3.44)$$

For each resulting reduced polarization state, the basis-dependent BBM92 error rates are evaluated and the operational metric is defined as

$$Q_{\text{worst}} = \max(Q_Z, Q_X). \quad (3.45)$$

The adopted operating condition is

$$\boxed{Q_{\text{worst}} < 0.11.} \quad (3.46)$$

3.4.5. Two-Arm Operating Regions

Figure 3.7 shows the resulting worst-basis QBER in the $(\Delta\tau_A, \Delta\tau_B)$ plane for the three representative spectral-correlation conditions.

The three panels show that spectral correlation changes not only the magnitude of PMD-induced degradation but also the geometry of the two-arm operating region.

For $r = 0$, the frequency offsets of the two photons are statistically uncorrelated. Increasing the DGD in either arm therefore progressively increases the effective polarization degradation, and the BBM92 operating region remains concentrated around configurations in which the two DGDs are sufficiently small.

For $r = +0.9$, the two frequency offsets tend to have the same sign. For the aligned PMD configuration considered here, the corresponding frequency-dependent contributions tend

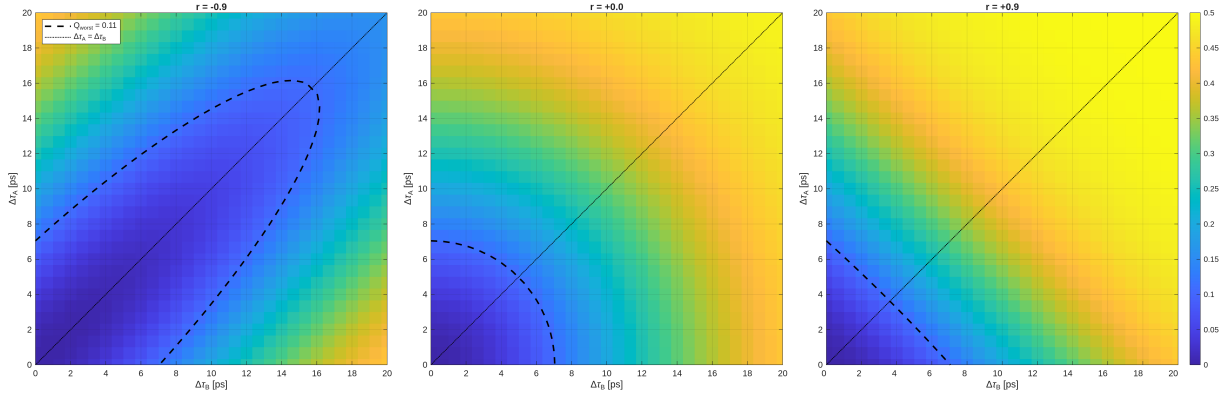


Figure 3.7: Worst-basis BBM92 QBER as a function of the differential group delays $\Delta\tau_A$ and $\Delta\tau_B$ for strongly anticorrelated ($r = -0.9$), uncorrelated ($r = 0$), and strongly positively correlated ($r = +0.9$) photon spectra. The dashed contour identifies the adopted operating boundary $Q_{\text{worst}} = 0.11$, while the dotted line denotes the symmetric condition $\Delta\tau_A = \Delta\tau_B$.

to reinforce each other. The operating region is consequently reduced when both arms simultaneously exhibit significant DGD.

A qualitatively different behavior is observed for $r = -0.9$. The frequency offsets now preferentially have opposite signs. When the DGDs are comparable, the two PMD-induced contributions can partially compensate. The BBM92 operating region therefore extends toward substantially larger DGDs in the vicinity of

$$\Delta\tau_A \simeq \Delta\tau_B. \quad (3.47)$$

The diagonal extension visible in the left panel is therefore a direct consequence of the interaction between the anticorrelated source spectrum and the frequency-dependent transformations of the two fiber arms. In particular, large individual DGDs do not necessarily imply a correspondingly large polarization degradation when their effects are correlated through the joint spectrum.

3.4.6. Spectral Correlation and Symmetric-DGD Tolerance

The influence of spectral correlation is quantified more directly by restricting the two-arm parameter space to

$$\Delta\tau_A = \Delta\tau_B = \Delta\tau \quad (3.48)$$

and determining, for every value of r , the maximum DGD compatible with the adopted BBM92 criterion.

The resulting threshold $\Delta\tau_{\text{th}}$ is shown in Fig. 3.8.

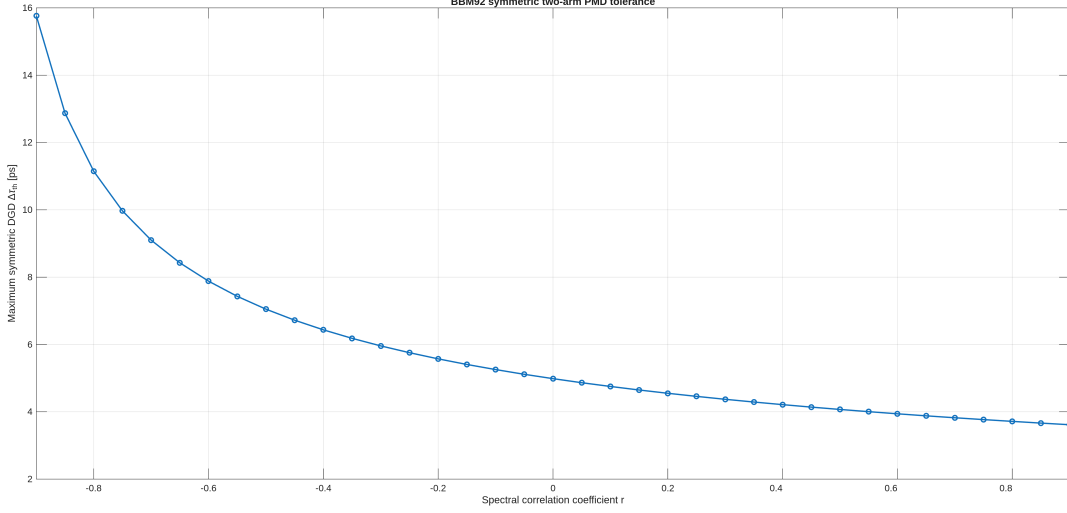


Figure 3.8: Maximum symmetric differential group delay compatible with the adopted BBM92 operating condition as a function of the spectral correlation coefficient r , for $\sigma_\omega = 1.0 \times 10^{11}$ rad/s. Negative spectral correlation substantially increases the tolerable symmetric DGD because the anticorrelated frequency offsets partially compensate the PMD contributions accumulated in the two aligned fiber arms.

The numerical boundary varies monotonically with the spectral correlation coefficient. For the three representative conditions, the extracted threshold DGDs are

$$\begin{aligned}
 r = -0.9 : \quad & \Delta\tau_{\text{th}} = 15.763 \text{ ps}, \\
 r = 0 : \quad & \Delta\tau_{\text{th}} = 4.984 \text{ ps}, \\
 r = +0.9 : \quad & \Delta\tau_{\text{th}} = 3.615 \text{ ps}.
 \end{aligned} \tag{3.49}$$

For the fixed source bandwidth considered here, the corresponding normalized parameters

$$\chi_{\text{th}} = \sigma_\omega \Delta\tau_{\text{th}} \tag{3.50}$$

are

$$\chi_{\text{th}} = 1.5763, \quad 0.4984, \quad 0.3615, \tag{3.51}$$

for $r = -0.9$, $r = 0$, and $r = +0.9$, respectively.

Moving from strong positive spectral correlation to strong spectral anticorrelation therefore increases the maximum tolerable symmetric DGD from

$$3.615 \text{ ps} \quad \text{to} \quad 15.763 \text{ ps}, \quad (3.52)$$

corresponding to an increase by more than a factor of four.

This improvement is obtained without changing either the marginal source bandwidth or the individual PMD model of the two fibers. It originates entirely from the different joint spectral statistics of the two photons and their interaction with the frequency-dependent transformations accumulated in the two arms.

3.4.7. Operational Interpretation

The concurrence evaluated at the numerically extracted BBM92 boundary is

$$C_{\text{th}} \simeq 0.7800 \quad (3.53)$$

for all three representative spectral-correlation conditions.

This value is not imposed as an entanglement threshold. The operating boundary is determined exclusively from

$$Q_{\text{worst}} = 0.11, \quad (3.54)$$

and the concurrence is evaluated only after the corresponding physical configuration has been identified. The approximately constant value $C \simeq 0.78$ is therefore a property of the restricted aligned-axis Gaussian state family considered here and must not be interpreted as a universal concurrence requirement for BBM92 operation.

The main result of the experiment is instead that two-arm PMD cannot generally be characterized by the individual DGDs alone. The same pair of fiber parameters can produce substantially different polarization degradation depending on the spectral correlation of the entangled-photon source.

The physical-to-operational mapping investigated in this experiment can therefore be summarized as

$$\boxed{(\Delta\tau_A, \Delta\tau_B, r) \longrightarrow \rho_{\text{pol}} \longrightarrow Q_{\text{worst}} \longrightarrow \mathcal{R}_{\text{BBM92}}.} \quad (3.55)$$

Spectral correlation thus becomes an additional source parameter controlling the robustness of the distributed polarization state against two-arm PMD. In the specific aligned configuration considered here, strong spectral anticorrelation can substantially enlarge the BBM92 operating region when the differential group delays of the two arms are appropriately matched.

3.5. Compensation Robustness and Operational Tolerances

The previous experiments characterize the BBM92 operating region under idealized and controlled PMD configurations. In a practical compensated link, however, the parameters of the compensating transformation cannot in general be expected to match the propagation channel exactly.

This experiment will therefore investigate the robustness of the BBM92 operating region against imperfect PMD compensation. In particular, the analysis will consider errors in the compensation DGD and misalignment between the relevant polarization axes. The objective will be to translate these imperfections into operational tolerances by determining the combinations of compensation errors for which

$$Q_{\text{worst}} < Q_{\text{th}}. \quad (3.56)$$

The resulting maps will provide a direct measure of how accurately the compensation parameters must reproduce the fiber transformation in order to maintain the link inside the adopted BBM92 operating region.

3.6. Dependence of the Operating Region on the Fiber Model

The effective polarization channels introduced in Chapter 2 provide compact descriptions of dephasing and depolarization, while the PMD model explicitly includes the frequency dependence of birefringent propagation and the subsequent spectral trace.

This experiment will compare the BBM92 operating regions predicted by these different

levels of description. The objective will not be to require the effective and frequency-dependent models to be physically identical, but rather to determine under which conditions a reduced polarization-only model provides an adequate approximation of the operational degradation predicted by the more detailed PMD description.

The comparison will therefore focus on the displacement of the $Q_{\text{worst}} = Q_{\text{th}}$ boundary and will identify parameter regimes in which the simplified description reproduces the relevant BBM92 operating region, as well as regimes in which the explicit spectral dependence of the fiber can no longer be neglected.

3.7. Combined BBM92 Fiber Operating Maps

The final numerical experiment will combine the principal physical dependencies identified throughout the preceding analyses into a unified operational description of the optical link.

Rather than isolating one propagation mechanism at a time, selected source, fiber, and compensation parameters will be varied jointly in order to construct BBM92 operating maps representative of the complete modeling framework developed in this thesis. The objective will be to identify the regions of the resulting multidimensional parameter space satisfying

$$\xi \in \mathcal{R}_{\text{BBM92}}, \quad (3.57)$$

and to determine which physical parameters most strongly constrain the boundary of this region.

This final analysis will therefore synthesize the results of the preceding experiments into a set of source–fiber operating conditions, providing the numerical connection between the physical parameters of the modeled optical link and its predicted suitability for polarization-entanglement distribution under the adopted BBM92 criterion.

4 | Experimental Activity for Polarization Control

This chapter presents the experimental activity carried out at the POLICOM laboratory of the Department of Electronics, Information and Bioengineering (DEIB) at Politecnico di Milano.

4.1. Experimental Objectives and Development Strategy

The main objective of the experimental work is to investigate, characterize, and control the polarization behavior of light transmitted through a two-arm fiber-optic system, with particular attention to the mechanisms that may affect the transmission of polarization-encoded quantum states. The experimental activity therefore complements the analytical and numerical framework developed in the previous chapters by providing a laboratory-based study of the same physical phenomena.

The adopted architecture is intentionally simplified. Rather than implementing a complete quantum-communication protocol, the setup reproduces a representative subsystem of a future entanglement-based BBM92 architecture. In this context, the objective is to study the behavior of the optical components and polarization-control stages that would be required in a realistic fiber-based installation, and to identify the experimentally accessible observables that can be used to monitor their operation.

The experimental platform combines several optical and electronic components previously characterized and calibrated in the laboratory, including a spontaneous-parametric-down-conversion source, polarization-control elements, polarization beam splitters, polarization-maintaining fiber sections, single-photon avalanche detectors, time-tagging electronics, and, in the later development stages, an electronically controlled polarization controller. These components are progressively integrated into a common measurement and control framework.

The experimental activity was developed according to three main stages:

1. **Manual polarization control.** The first stage establishes the basic two-arm detection architecture using four single-photon avalanche detectors and manually adjusted polarization control. This configuration provides the reference measurement platform for the characterization of single-photon count rates and coincidence observables.
2. **Automatic polarization control.** The second stage introduces electronic polarization control and the corresponding software-based feedback system. The controller is first calibrated and then used to investigate automatic polarization alignment. Different feedback and optimization strategies are tested in order to evaluate their ability to recover and maintain a desired polarization condition from experimentally measured coincidence observables.
3. **Controlled PM-fiber propagation experiment.** The final stage combines the four-detector analysis, automatic polarization control, and polarization-maintaining (PM) fiber sections in a single experimental setup. The PM fiber is used to introduce a controlled birefringent propagation mechanism and to investigate the resulting polarization degradation under experimentally reproducible conditions.

The final PM-fiber experiment is performed using a non-entangled input polarization state. Consequently, it does not directly measure entanglement degradation, but it allows the polarization-decoherence and depolarization mechanisms induced by controlled birefringent propagation to be experimentally characterized. The measured coincidence observables are then interpreted through the calibrated physical PM-fiber model developed in the previous chapters.

More generally, the scope of the experimental activity is not to reconstruct the microscopic evolution of each individual photon inside the fiber. The objective is instead to determine how the available optical components, polarization controllers, detectors, and fiber sections behave when assembled into a realistic quantum-communication link. In this sense, the laboratory platform provides an engineering-level validation environment connecting the theoretical fiber models with experimentally accessible quantities and with the requirements of a future BBM92-oriented implementation.

4.2. Experimental Platform and Instrumentation

This section describes the common hardware and acquisition infrastructure used throughout the experimental activity.

The presentation includes the SPDC source and the optical architecture inherited from the previously developed laboratory platform, manual polarization controllers, the electronic polarization controller, polarizing beam splitters, polarization-maintaining fibers, SPAD detectors, and the time-tagging system.

The software developed for detector configuration, real-time singles monitoring, coincidence-histogram acquisition, timing-delay compensation, EPC control, and experimental data storage is also introduced.

The description focuses on the role played by each component in the experiments performed in this work, while detailed characterization of instrumentation already documented in previous laboratory work is only recalled where necessary.

4.3. Phase I: Manual Polarization Control and Four-SPAD Measurements

The first experimental phase establishes the reference measurement configuration using four SPAD detectors and two manually operated polarization controllers.

The purpose of this stage is to validate the complete acquisition chain, including single-photon count-rate measurements, coincidence-peak identification, timing alignment between detector channels, and polarization-dependent redistribution of coincidence events.

Manual polarization control is used to align the polarization states with the measurement basis and to investigate how changes in the optical configuration modify the observed coincidence channels.

This phase provides the experimental baseline required before introducing automatic polarization compensation.

4.4. Phase II: Automatic Electronic Polarization Control

The second phase introduces an electronic polarization controller into a reduced experimental configuration without PM fiber.

One photon provides the reference detection event, while the polarization of the second optical arm is controlled by the EPC before being analyzed by a PBS and two SPAD detectors.

The polarization-compensation problem is formulated as a closed-loop, derivative-free optimization problem in which the EPC control voltages are updated according to experimentally measured coincidence observables.

This section introduces the feedback architecture, the EPC control variables, the measurement cycle, and the general optimization framework used throughout the automatic-control experiments.

4.5. Coincidence-Based Objective Functions

This section defines the experimental quantities used to provide feedback to the electronic polarization controller.

The principal observables are the net coincidence counts associated with the two outputs of the polarization analyzer. These quantities are extracted from the measured correlation histograms through coincidence-peak localization, integration, and background estimation.

The main objective functions used in the experiments are introduced and compared, including optimization of a selected coincidence channel and combined objective functions constructed from both polarization outputs.

Particular attention is given to the physical meaning of the objective functions and to the distinction between single-photon count rates and coincidence-based feedback.

4.6. EPC Optimization Algorithms

This section presents the principal optimization algorithms implemented for automatic EPC control.

The first method is sequential coordinate search, in which the EPC control parameters are optimized one at a time while retaining configurations that improve the measured objective function.

The second method is a stochastic coarse-to-fine optimization based on simulated annealing. A coarse exploration of the EPC voltage space is followed by a finer local search, while probabilistic acceptance of non-improving configurations provides a mechanism for escaping unfavorable local solutions.

The algorithms are discussed in terms of convergence behavior, measurement cost, repeatability, sensitivity to photon-counting fluctuations, and suitability for closed-loop

polarization control.

4.7. Phase III: Controlled Propagation Through PM Fiber

The final experimental phase introduces a controlled polarization-maintaining fiber section into the optical path.

The experimental configuration is

$$\text{SPDC OUT 1} \rightarrow \text{Manual PC} \rightarrow \text{PBS}_{45^\circ} \rightarrow \text{PM}_{45^\circ} \rightarrow [\text{PM}_{0^\circ}]^N \rightarrow \text{EPC} \rightarrow \text{PBS} \rightarrow \{\text{SPAD 3, S} \quad (4.1)$$

while the reference arm is

$$\text{SPDC OUT 2} \rightarrow \text{SPAD 1}. \quad (4.2)$$

Unlike the entanglement-distribution models developed in the theoretical part of the thesis, this experiment uses a non-entangled input polarization state.

Its purpose is to experimentally isolate and characterize the effect of controlled birefringent propagation and differential group delay on the measured polarization observables.

4.8. Role of the Rotated and Aligned PM-Fiber Sections

This section explains the physical motivation for the sequence composed of an initial PM-fiber section rotated by approximately 45° , followed by additional aligned PM-fiber sections.

The rotated section couples the prepared input polarization to both principal polarization modes of the PM fiber, allowing the fast and slow components to propagate simultaneously.

The subsequent aligned PM-fiber sections increase the accumulated differential group delay while preserving, to first approximation, the same principal-axis orientation.

The number of aligned sections, or equivalently the total PM-fiber length, is therefore used

as the main experimental control parameter governing the strength of the polarization–temporal separation.

4.9. Measured Coincidence Observables Under PM-Fiber Propagation

This section presents the quantities directly accessible with the current experimental instrumentation.

Since the available setup performs polarization analysis in a single measurement basis, full quantum-state tomography is not carried out. Consequently, quantities such as the complete density operator, concurrence, and state fidelity are not directly reconstructed from the experimental measurements.

The experimental analysis instead focuses on the coincidence observables associated with the two PBS output channels and on their evolution as the PM-fiber configuration is modified.

The measurements are used to quantify the redistribution and loss of polarization contrast produced by increasing differential propagation between the principal fiber modes.

4.10. Comparison with a Calibrated Physical PM-Fiber Model

The experimental results are interpreted through a calibrated physical model of the PM-fiber propagation rather than through complete quantum-state tomography.

The model incorporates the relevant properties of the PM-fiber sections, such as their length, principal-axis orientation, birefringence, differential group delay, and the spectral characteristics of the propagating photons.

From these parameters, the expected polarization evolution and the corresponding measurement probabilities at the PBS outputs are calculated and compared with the experimentally measured coincidence observables.

This model-assisted approach allows the physical origin of the observed polarization degradation to be investigated within the limitations imposed by the available measurement basis.

4.11. Connection with the Theoretical and Numerical Framework

The final section connects the experimental observations with the physical fiber-propagation framework developed in the previous chapters.

The PM-fiber experiment provides a controlled realization of frequency-dependent birefringent propagation, in which polarization becomes coupled to temporal or spectral degrees of freedom through differential group delay.

When these additional degrees of freedom are not resolved by the measurement, the resulting polarization dynamics can be represented through an effective reduced-state description.

The experiment therefore provides a physical test of the propagation mechanism underlying the numerical PMD model. The distinction between theory and experiment is maintained explicitly: the numerical framework treats both single-photon and entangled two-photon states, whereas the final laboratory experiment investigates the same physical mechanism using a non-entangled input state and a restricted set of coincidence observables.

Bibliography

- [1] Stephanie Wehner, David Elkouss, and Ronald Hanson. Quantum internet: A vision for the road ahead. *Science*, 362(6412):eaam9288, 2018. doi: 10.1126/science.aam9288.
- [2] Sergi Abadal, Chong Han, Vitaly Petrov, Laura Galluccio, Ian F. Akyildiz, and Josep M. Jornet. Electromagnetic nanonetworks beyond 6g: From wearable and implantable networks to on-chip and quantum communication. *IEEE Journal on Selected Areas in Communications*, 42(8):2122–2142, 2024. doi: 10.1109/JSAC.2024.3399253.
- [3] Charles H. Bennett, Gilles Brassard, and N. David Mermin. Quantum cryptography without Bell’s theorem. *Physical Review Letters*, 68(5):557–559, 1992. doi: 10.1103/PhysRevLett.68.557.
- [4] Evan Dowling, Mark Morris, Gerald Baumgartner, Rajarshi Roy, and Thomas E. Murphy. Non-local polarization alignment and control in fibers using feedback from correlated measurements of entangled photons. *Optics Express*, 31(2):2316–2329, 2023. doi: 10.1364/OE.475465.
- [5] Phoenix S. Y. Poon and C. K. Law. Polarization and frequency disentanglement of photons via stochastic polarization mode dispersion. *Physical Review A*, 77(3):032327, 2008. doi: 10.1103/PhysRevA.77.032327.
- [6] Misha Brodsky, Elizabeth C. George, Cristian Antonelli, and Mark Shtaif. Loss of polarization entanglement in a fiber-optic system with polarization mode dispersion in one optical path. *Optics Express*, 19(1):45–54, 2011. doi: 10.1364/OE.19.000045.
- [7] Gabriele Riccardi. *Polarization-Entanglement Distribution in Fiber-Optic Channels*. PhD thesis, Università degli Studi dell’Aquila, L’Aquila, Italy, 2021. Doctoral Program in Physical and Chemical Sciences, XXXIII cycle, academic year 2019/2020.
- [8] P. K. A. Wai and C. R. Menyuk. Polarization mode dispersion, decorrelation, and diffusion in optical fibers with randomly varying birefringence. *Journal of Lightwave Technology*, 14(2):148–157, 1996. doi: 10.1109/50.482256.

- [9] Cristian Antonelli, Mark Shtaif, and Misha Brodsky. Sudden death of entanglement induced by polarization mode dispersion. *Physical Review Letters*, 106(8):080404, 2011. doi: 10.1103/PhysRevLett.106.080404.
- [10] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, Cambridge, 10th anniversary edition edition, 2010. ISBN 9781107002173.
- [11] Karl Kraus. General state changes in quantum theory. *Annals of Physics*, 64(2): 311–335, 1971. doi: 10.1016/0003-4916(71)90108-4.
- [12] Joseph B. Altepeter, Daniel F. V. James, and Paul G. Kwiat. Quantum state tomography. In Matteo G. A. Paris and Jaroslav Řeháček, editors, *Quantum State Estimation*, pages 113–145. Springer, Berlin, Heidelberg, 2004. doi: 10.1007/978-3-540-44481-7_4.
- [13] Richard Jozsa. Fidelity for mixed quantum states. *Journal of Modern Optics*, 41(12):2315–2323, 1994.
- [14] William K. Wootters. Entanglement of formation of an arbitrary state of two qubits. *Physical Review Letters*, 80(10):2245–2248, 1998. doi: 10.1103/PhysRevLett.80.2245.
- [15] Di Nardo C. Entanglement-based quantum key distribution systems, 2026.
- [16] Warren P. Grice and Ian A. Walmsley. Spectral information and distinguishability in type-II down-conversion with a broadband pump. *Physical Review A*, 56(2):1627–1634, 1997. doi: 10.1103/PhysRevA.56.1627.
- [17] J. P. Gordon and H. Kogelnik. PMD fundamentals: Polarization mode dispersion in optical fibers. *Proceedings of the National Academy of Sciences of the United States of America*, 97(9):4541–4550, 2000. doi: 10.1073/pnas.97.9.4541.

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